River Water Quality in Hong Kong in 2016

Environmental Protection Department
The Government of the Hong Kong Special Administrative Region

Mission

To conduct an effective and scientifically sound water monitoring programme that helps protect the health of Hong Kong's rivers and streams and ensure the water quality objectives can be achieved and maintained.

Lam Tsuen River

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Acknowledgement

We acknowledge the Government Laboratory for undertaking chemical analyses of the water samples.

Content

Appendices

1. Introduction

Hong Kong's rivers are generally relatively short with small flows, with their upstream sections mainly located within water gathering grounds and have been drawn on quite extensively for supplies of drinking water. At present, other than those located within water gathering grounds, the primary beneficial uses of our rivers include mainly maintenance of aquatic life, general amenity, flood prevention and storm water drainage. The main channel of Shing Mun River is the only watercourse currently used for secondary contact recreation.

The Environmental Protection Department (EPD) has initiated a comprehensive river water quality monitoring programme since 1986. It mainly monitors major rivers flowing through the urban areas, with one to several representative monitoring stations set up at the upstream and downstream areas of the main channel and tributaries. A number of small rivers and streams in the New Territories are also monitored for water pollution control and management purposes.

The monitoring programme serves the following purposes:

- evaluate the pollution status of river waters;
- monitor long-term changes in water quality;
- provide a scientific basis for planning water pollution control strategies; and
- assess the compliance with the key statutory Water Quality Objectives (WQOs).

This report summarises the water quality of rivers covered by EPD's river monitoring programme in 2016. Annual reports can be downloaded from the EPD's website:
http://wqrc.epd.gov.hk/en/water-quality/river-2.aspx

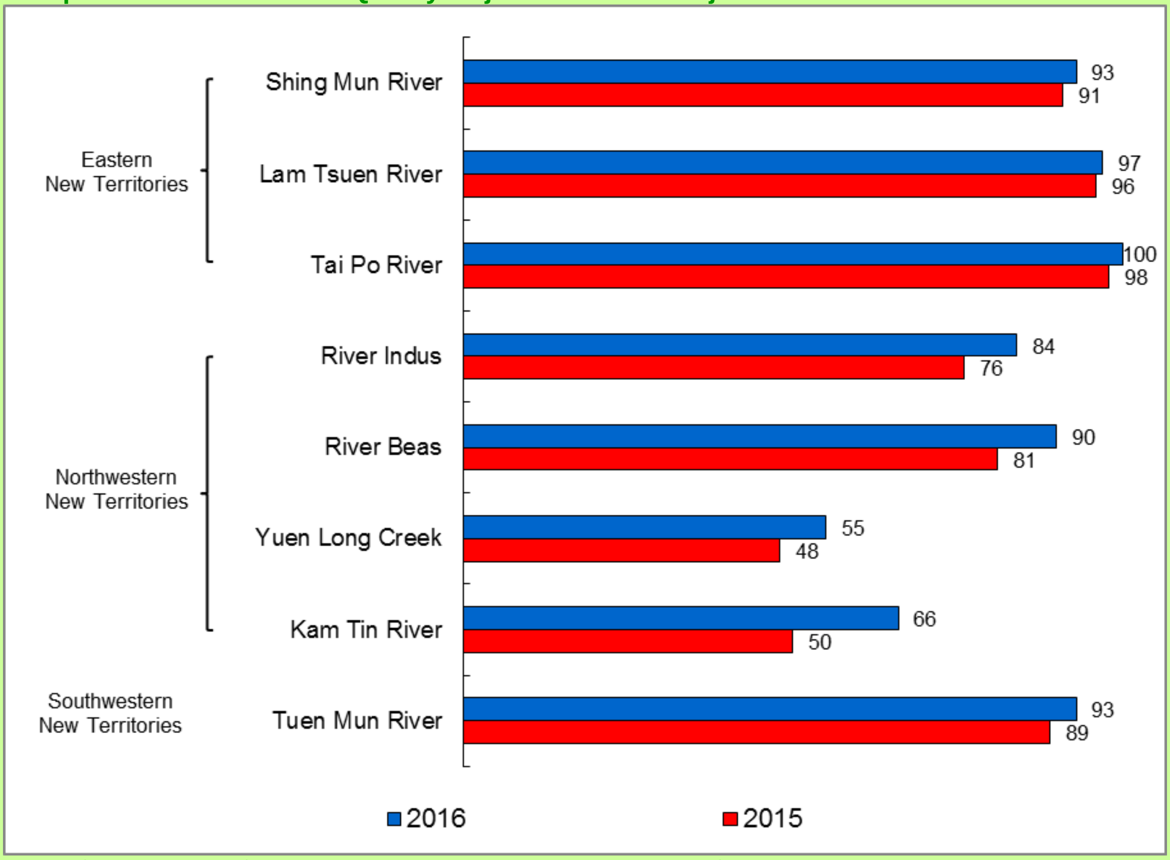
Field measurement at Yuen Long Creek

2. An overview of Hong Kong's rivers in 2016

In 2016, the EPD's river monitoring programme covered 30 rivers and a total of 82 monitoring stations, as compared with some 14 rivers and 47 stations when the programme started in 1986. The monitoring programme involved monthly conducting field measurements and collecting water samples for laboratory analyses. Over 40 physico-chemical and biological parameters, including organic matters, nutrients, metals and *E. coli* bacteria were analysed.

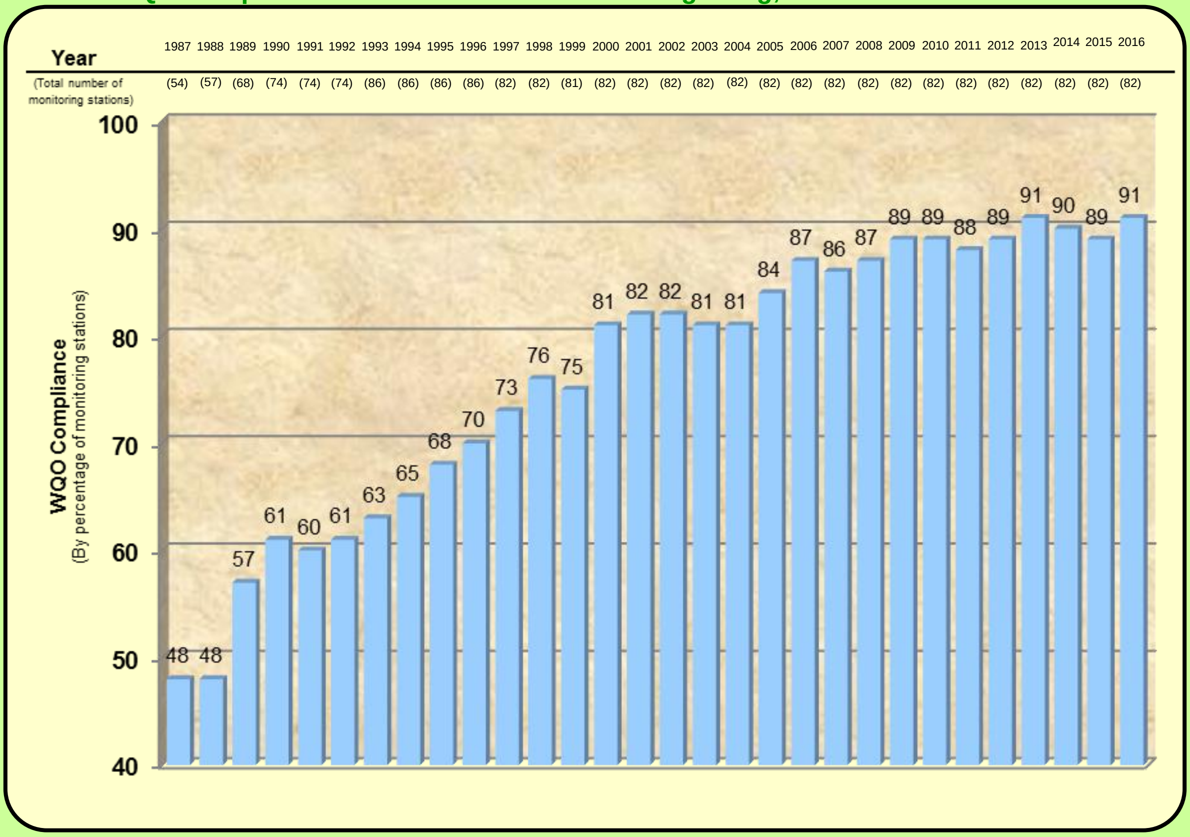
To assess compliance with the statutory Water Quality Objectives (WQOs), five representative parameters are used: pH, suspended solids, dissolved oxygen, 5-day biochemical oxygen demand, and chemical oxygen demand. In terms of the compliance rate, the overall water quality of Hong Kong rivers in 2016 continued to be good with an overall compliance rate at 91%, compared with 89% in 2015 and 90% in 2014. The high compliance rates in recent years were the result of implementation of pollution control legislation, including the Water Pollution Control Ordinance and the Livestock Waste Control Scheme introduced under the Waste Disposal Ordinance, as well as the extension of the sewer network under the Sewage Master Plans and the gradual connection of the village houses to public sewers.

Compliance with the Water Quality Objectives of the major rivers in 2016 and 2015



Please refer to Appendix G for water quality improvements and pollution loading of major rivers

Overall WQO compliance in the inland waters of Hong Kong, 1987-2016



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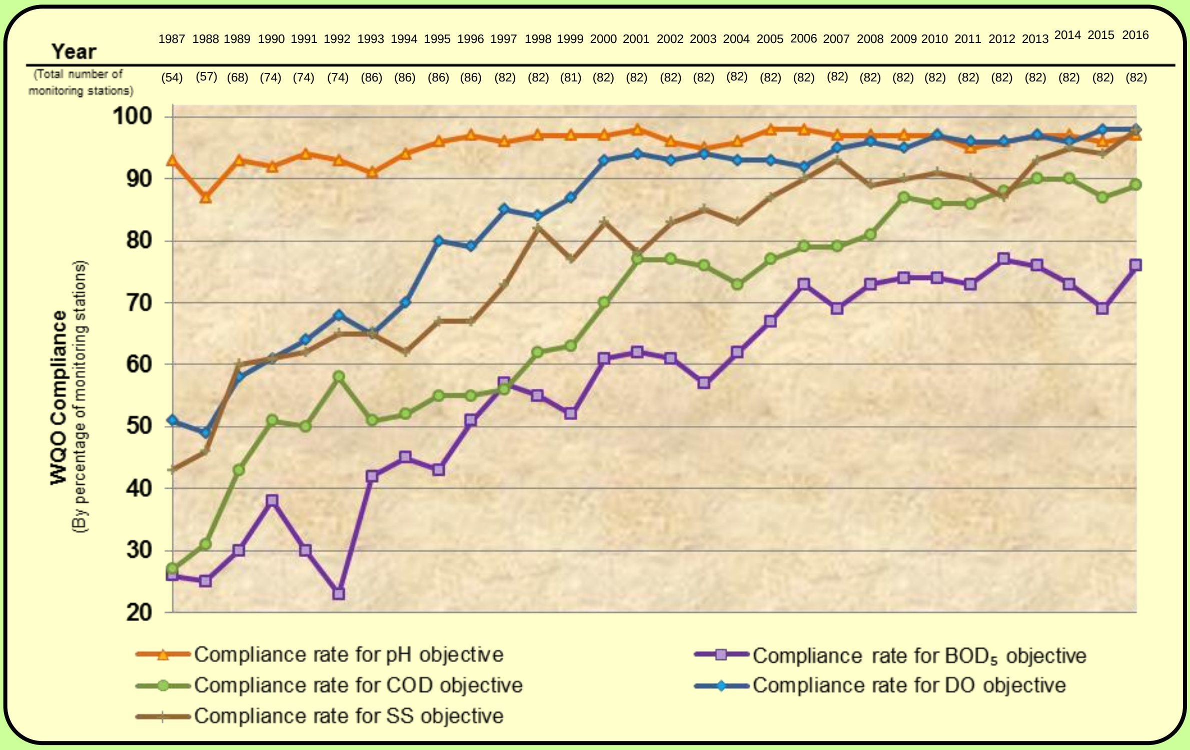
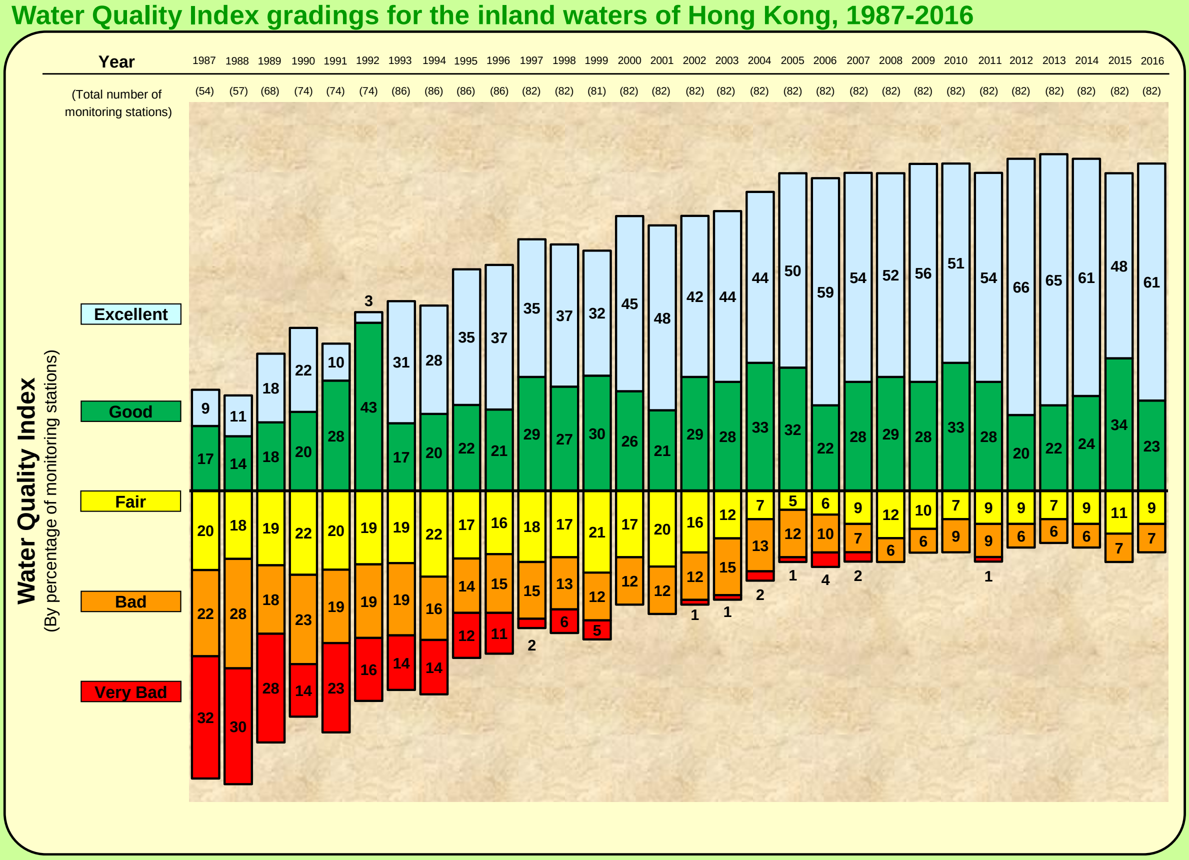
WQO compliance for five representative parameters in the inland waters of Hong Kong, 1987-2016



The Water Quality Index (WQI) is relevant to conserving the primary beneficial use of rivers for maintenance of aquatic life. It grades the general ecological health of rivers into five categories of “Excellent”, “Good”, “Fair”, “Bad” and “Very Bad” by assessing the three key water quality parameters: dissolved oxygen, 5-day biochemical oxygen demand and ammonia-nitrogen content[1]. As demonstrated by the WQI, there have been significant improvements in the river water quality in Hong Kong over the past three decades and continued to be good in 2016. Compared with 2015, 17 stations moved up a grade and four stations moved down a grade in WQI grading in 2016. These minor changes are considered to be within the normal range of natural fluctuations over the past ten years. In 2016, 84% of the river monitoring stations were graded “Good” or “Excellent”, as compared with 26% in 1987, suggesting that the river water quality has greatly improved and the pollution loadings in these watercourses were low. The majority of the monitoring stations in Lantau Island, the Eastern New Territories, Southwestern New Territories and Kowloon were in these two categories. On the other hand, only 7% of the monitoring stations were graded “Bad” and no station was graded “Very Bad” in 2016, as compared with 22% “Bad” and 32% “Very Bad” in 1987. Most of the “Bad” stations are located along watercourses in the Northwestern New Territories.

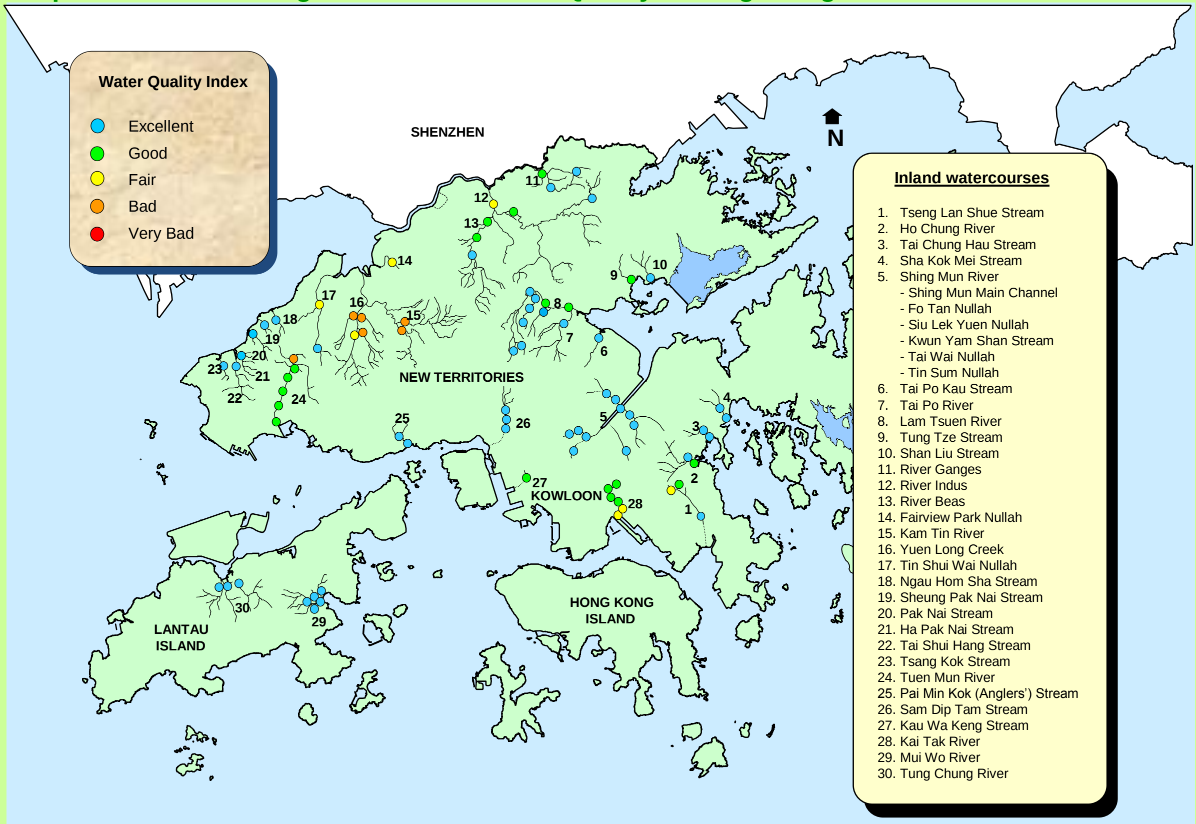
Water Quality Index gradings for the inland waters of Hong Kong, 1987-2016



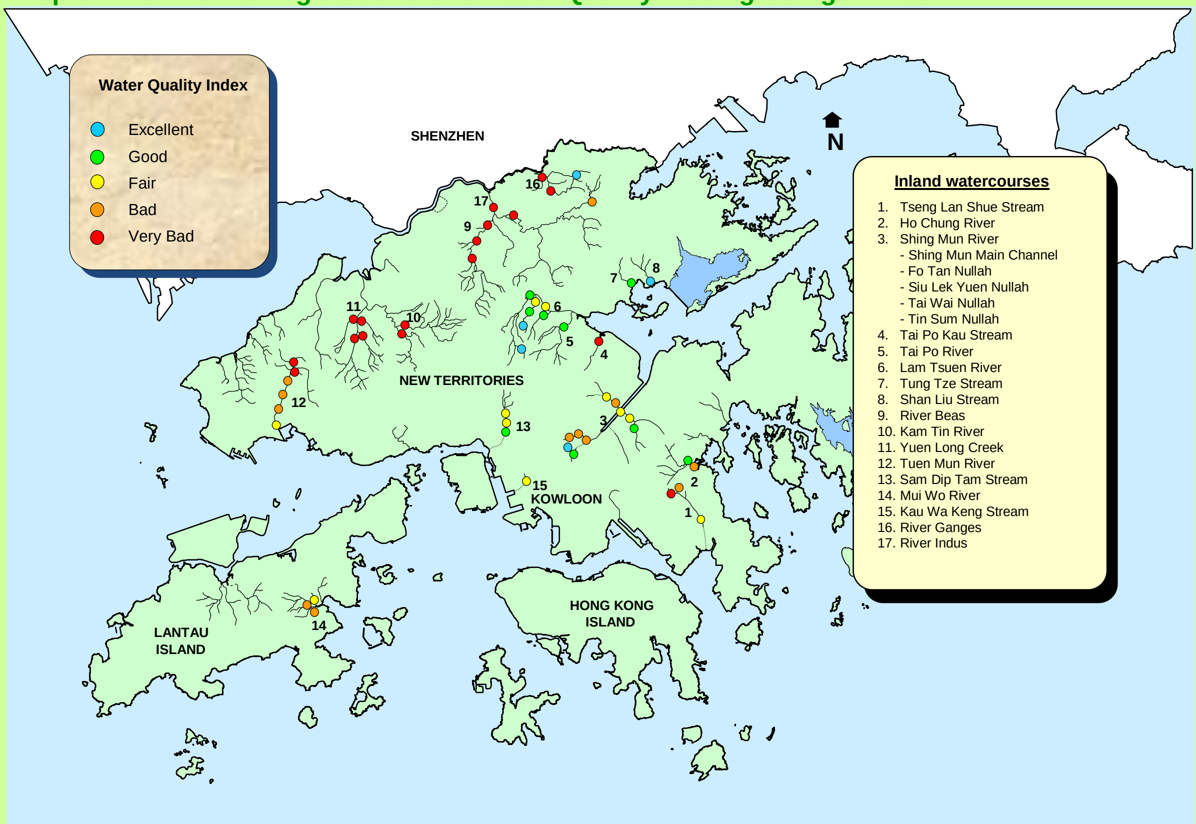
Figures are rounded to the nearest integer

[1] Please refer to Appendix F for more details on the calculation and assessment of Water Quality Index.

Map of river monitoring stations and Water Quality Index gradings in 2016



Map of river monitoring stations and Water Quality Index gradings in 1987

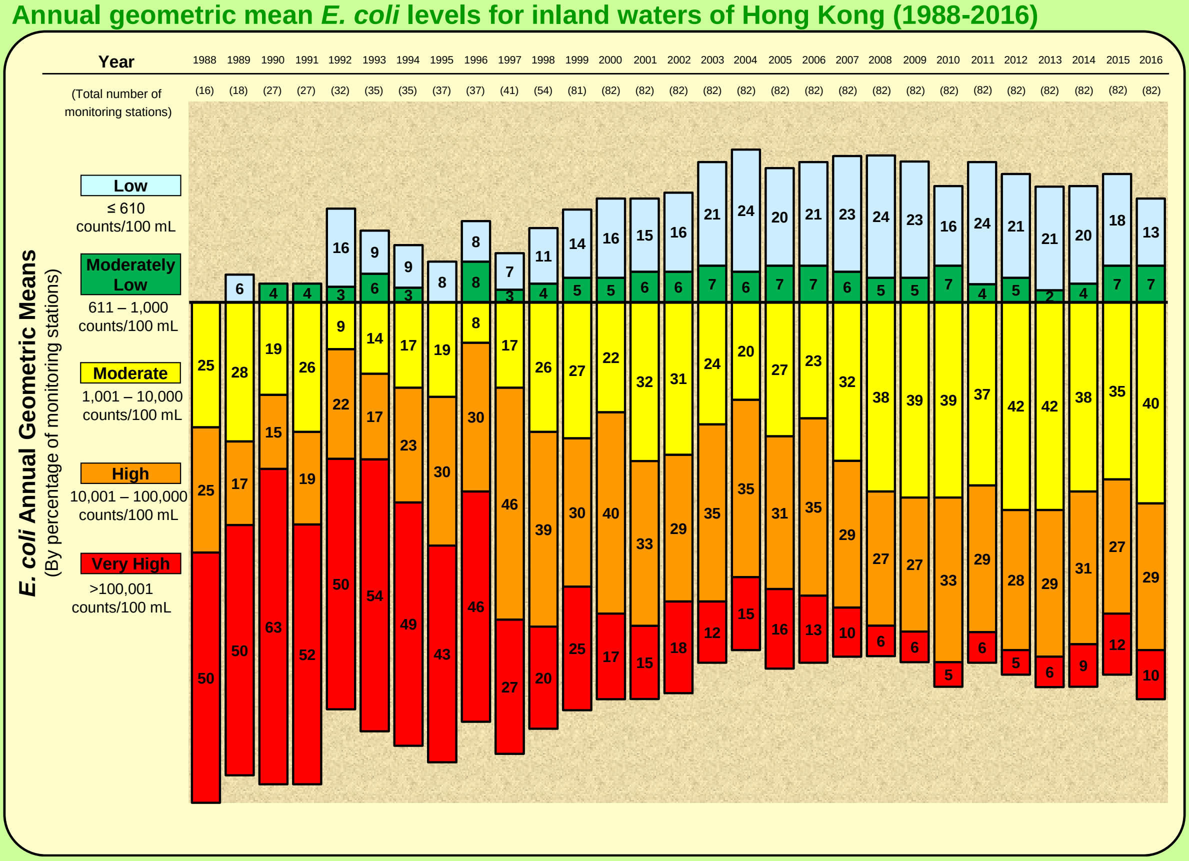


Since 1988, *E. coli* level in the river waters was also analysed as faeces of all warm-blooded animals contain *E. coli* and therefore monitoring the level of *E. coli* in a water body is a common indicator of the presence and extent of faecal contamination.

In 2016, 20% of our stations had "Low" or "Moderately Low" levels of *E. coli* (i.e. less than or equal to 1 000 counts/100mL) while 39% of our stations recorded "High" or "Very High" (i.e. over 10 000 counts/100mL) levels[2].

Most of the stations with "Very High" levels of *E. coli* are located in the Northwestern part of the New Territories (e.g. Yuen Long Creek, Kam Tin River and River Indus), mostly due to discharges from livestock farms, runoff from unsewered village houses as well as expedient connections in the old districts. While these watercourses are primarily intended for stormwater conveyance and flood prevention purpose, the Government will continue its efforts in enforcement and extension of the public sewer network to reduce the *E. coli* levels at these locations.

Annual geometric mean *E. coli* levels for inland waters of Hong Kong (1988-2016)



Figures are rounded to the nearest integer

In summary, Hong Kong's inland watercourses in 2016 were largely similar to 2015's monitoring results and showed long term improvement as a result of the gradual reduction of pollution loading in our rivers.

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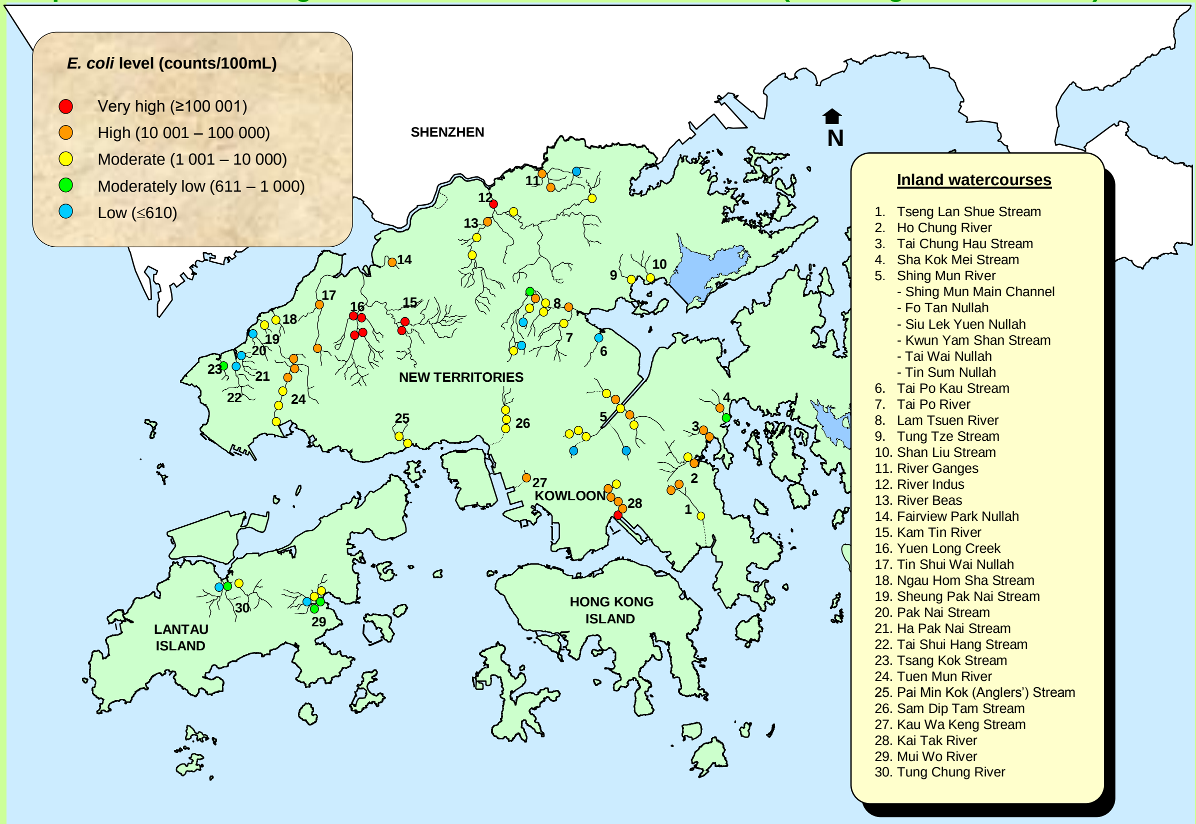
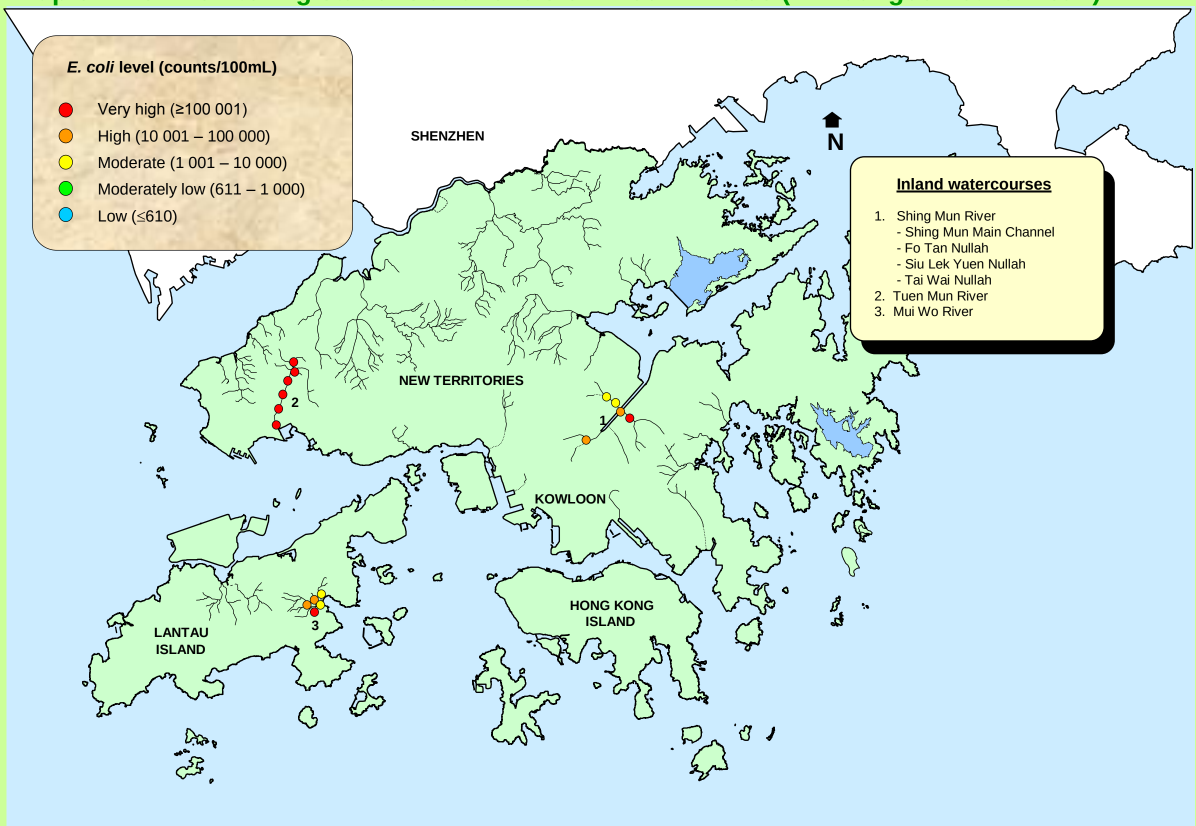
[2] All levels of *E. coli* in this report are reported as annual geometric means (counts/100mL).

Map of river monitoring stations and levels of *E. coli* in 2016 (Annual geometric mean)



Map of river monitoring stations and levels of *E. coli* in 1988 (Annual geometric mean)



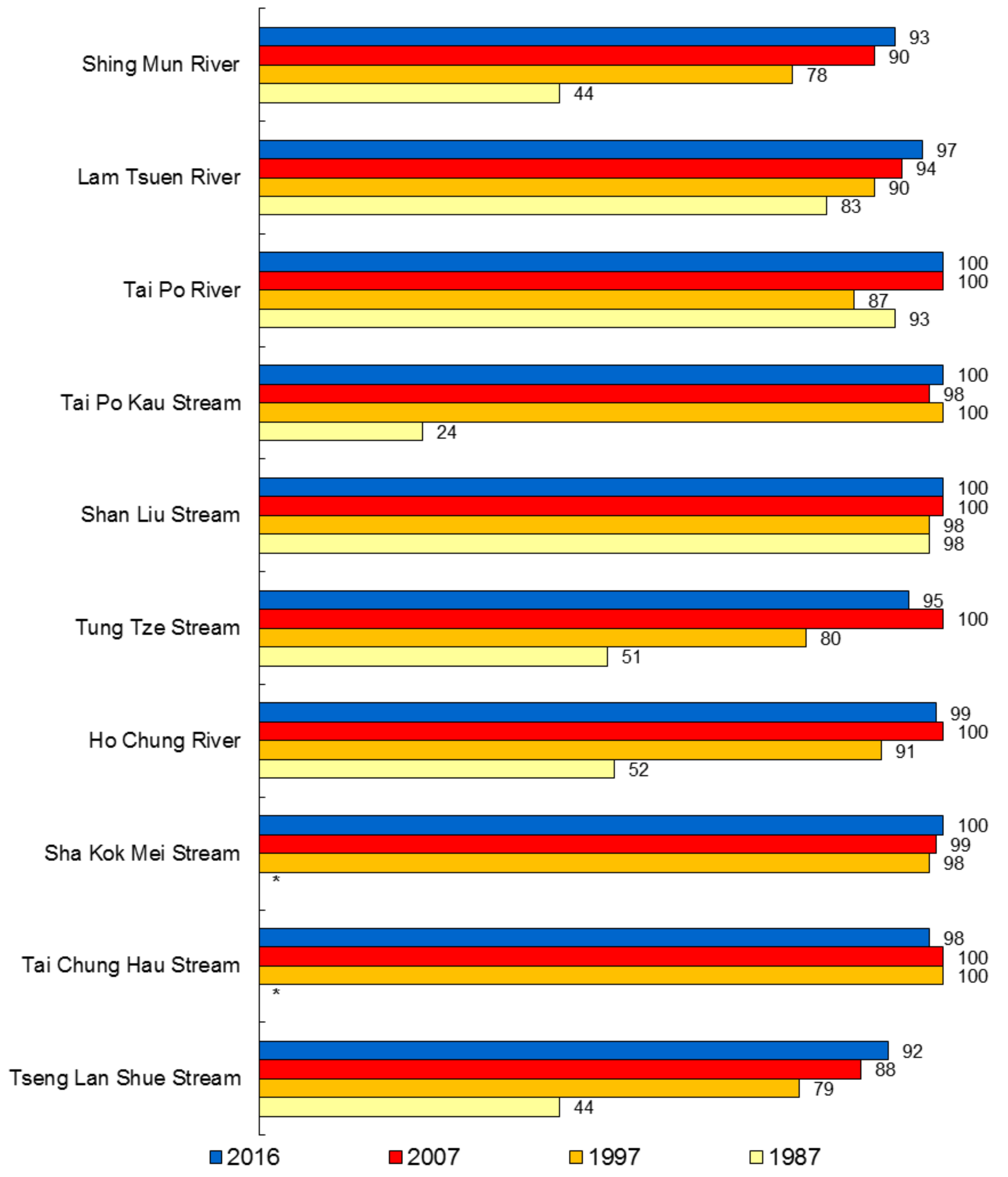
3. Eastern New Territories rivers

The EPD monitored ten watercourses in the Eastern New Territories in 2016. Six are located in the Tolo Harbour and Channel Water Control Zone (WCZ): Shing Mun River, Lam Tsuen River, Tai Po River, Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream. Within the Port Shelter WCZ, there are Ho Chung River, Sha Kok Mei Stream and Tai Chung Hau Stream. Tseng Lan Shue Stream is located within the Junk Bay WCZ.

The water quality of rivers and streams in the Eastern New Territories is good. In 2016, the overall WQO compliance rate was 97%, as in 2015. Within the region, four rivers achieved full compliance (100%) with the WQOs in 2016. These are Tai Po River, Tai Po Kau Stream and Shan Liu Stream in the Tolo Harbour and Channel WCZ and Sha Kok Mei Stream in the Port Shelter WCZ.

Shing Mun River

Compliance with the Water Quality Objectives of the rivers in the Eastern New Territories over the past three decades in percentage



* Monitoring data for Sha Kok Mei Stream and Tai Chung Hau Stream in 1987 were not available

Shing Mun River, a major river which has three main tributaries and runs through the densely populated Sha Tin urban area, showed marked improvement during the past three decades. The WQO compliance rate of Shing Mun River was 93% in 2016. Similarly, Lam Tsuen

River, as a major river which passes through the urban area of Tai Po and joins with Tai Po River before entering Tolo Harbour, recorded a WQO compliance rate of 97% in 2016.

Lam Tsuen River upstream

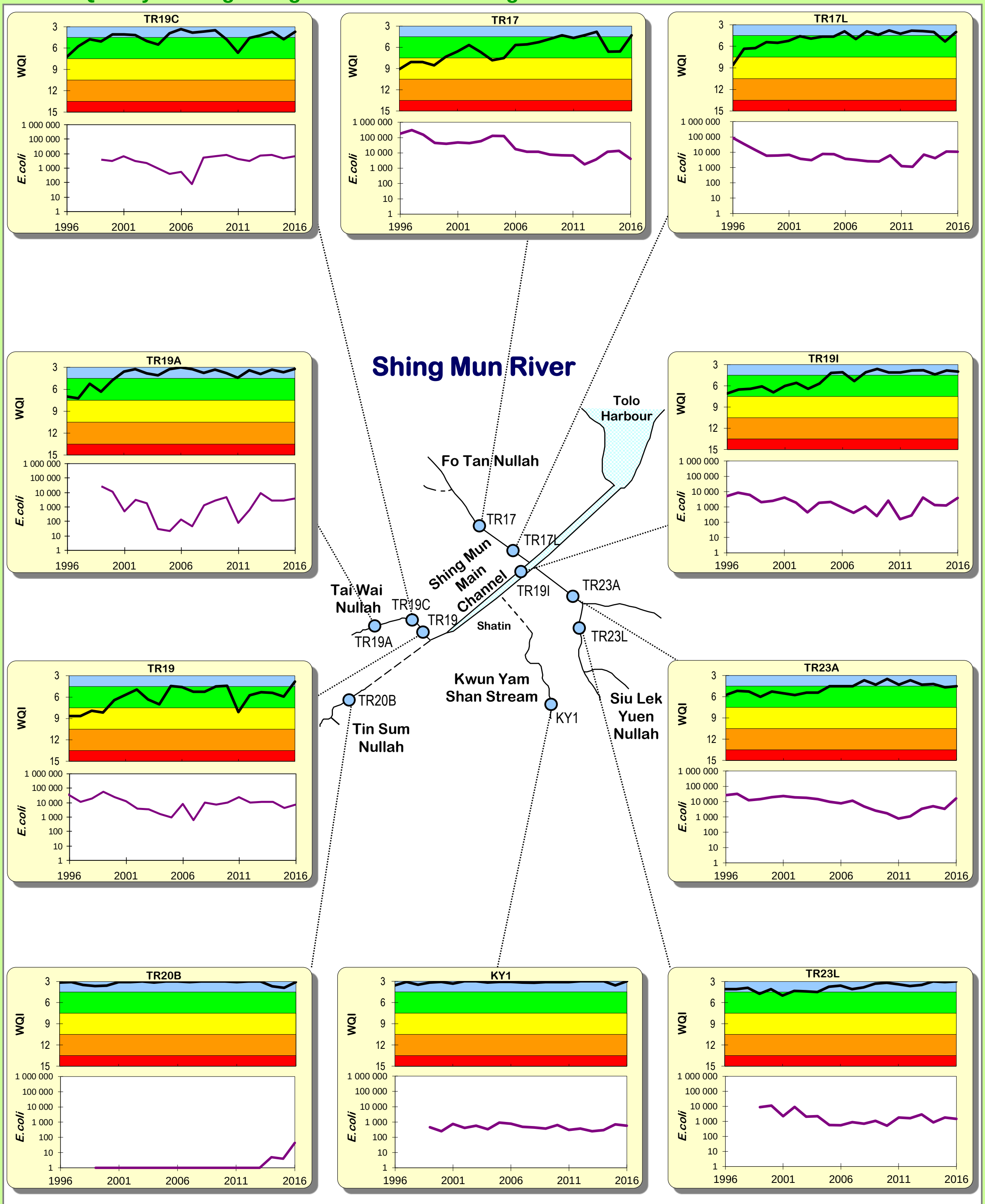
All the three small rivers in Port Shelter WCZ, namely Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream, achieved a WQO compliance rate of 98% or above in 2016.

Tseng Lan Shue Stream, recorded a compliance rate of 92% in 2016.

As for the WQI grading, 31 out of 32 (97%) of the river monitoring stations in the Eastern New Territories were graded "Good" or "Excellent" in 2016, same as in 2015. The remaining one station, i.e. Tseng Lan Shue Stream (JR3) at Tseng Lan Shue Tsuen, was in the "Fair" category. The overall improvement was due to the continued enforcement of the pollution control legislation, the implementation of Sewage Master Plans and extension of the public sewer to villages in the catchments.

The *E. coli* level at Shing Mun River Main Channel (TR19I) was considered as "Moderate" in 2016. While the slight deterioration as compared with 2015 was partly due to heavy rainfall as well as fewer bright sunshine hours in 2016, an action plan has already been put in place by EPD and DSD to inspect the sewer conditions in the catchment area.

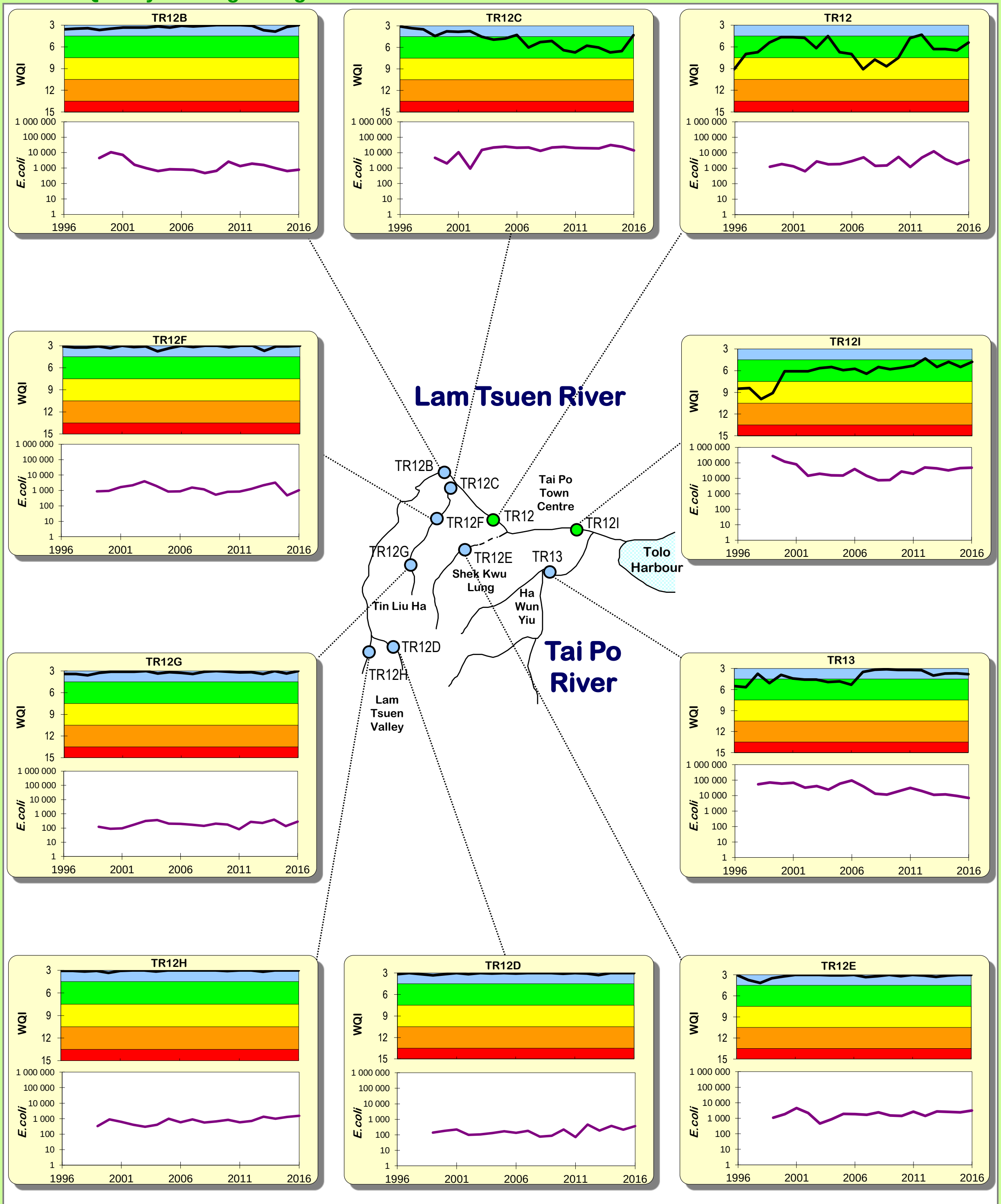
Water Quality Index gradings and *E. coli* in Shing Mun River



N.B.: WQI Annual Water Quality Index (Excellent, Good, Fair, Bad, Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

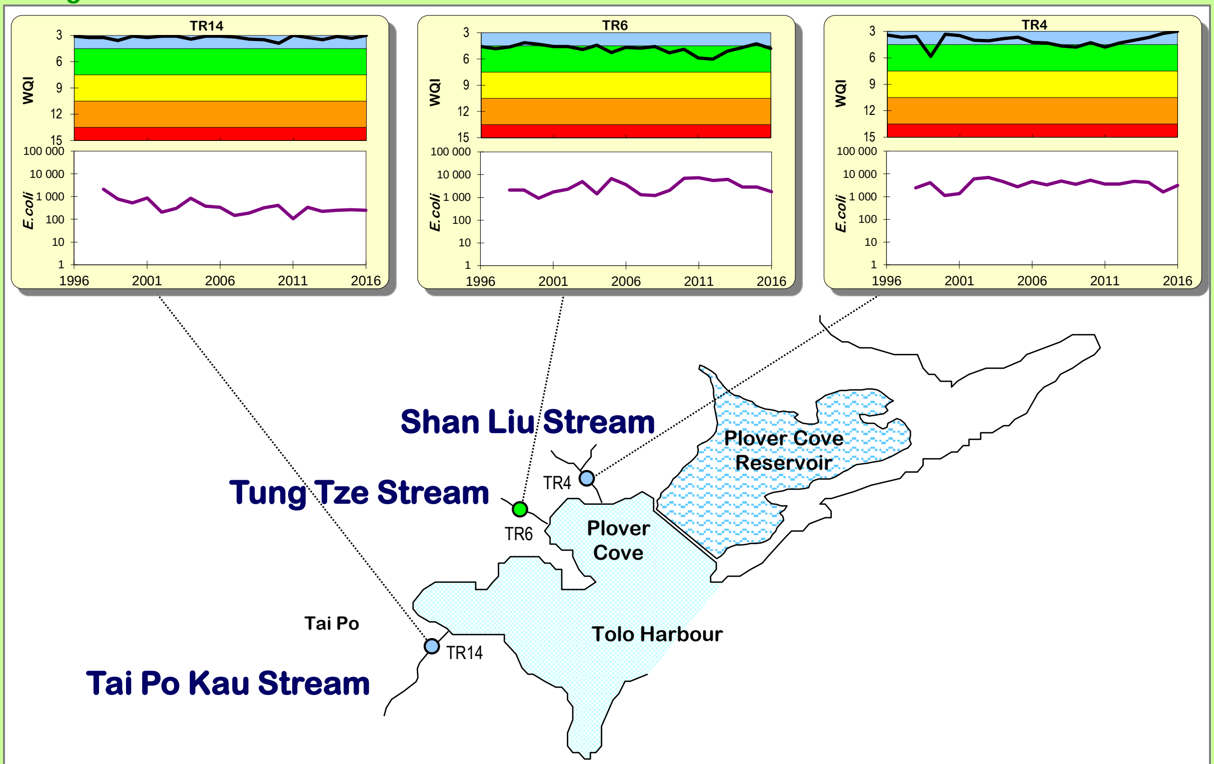
Water Quality Index gradings and *E. coli* levels in Lam Tsuen River and Tai Po River



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

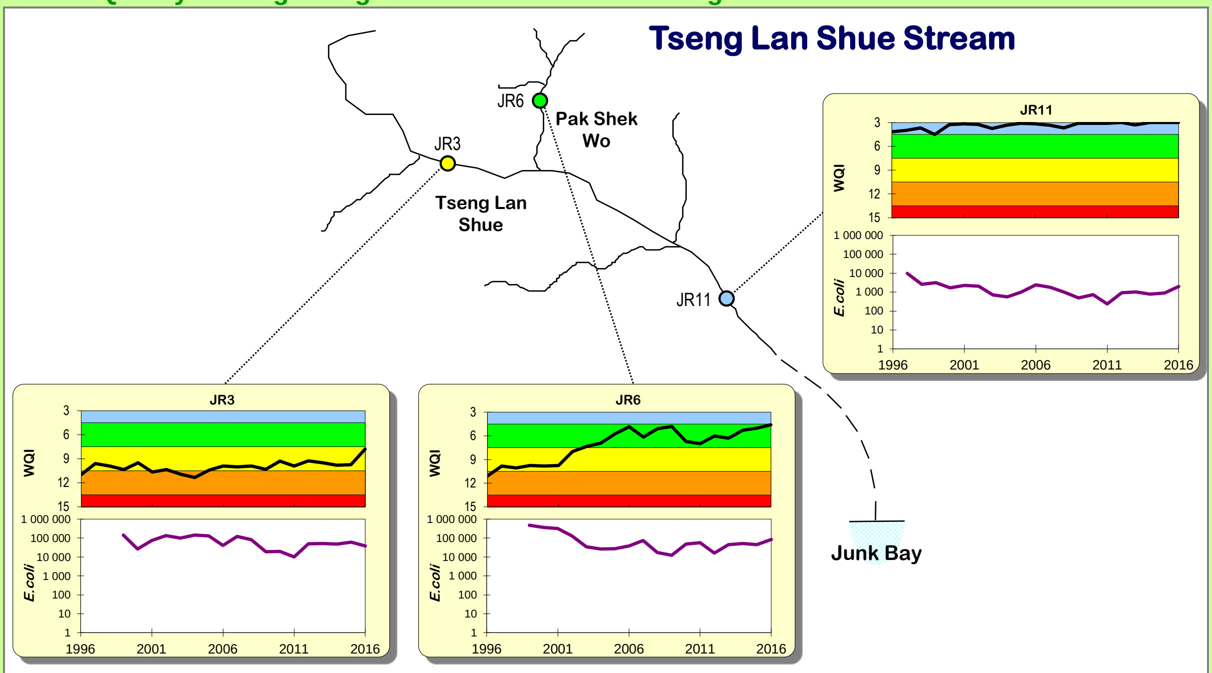
Water Quality Index gradings and *E. coli* levels in Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

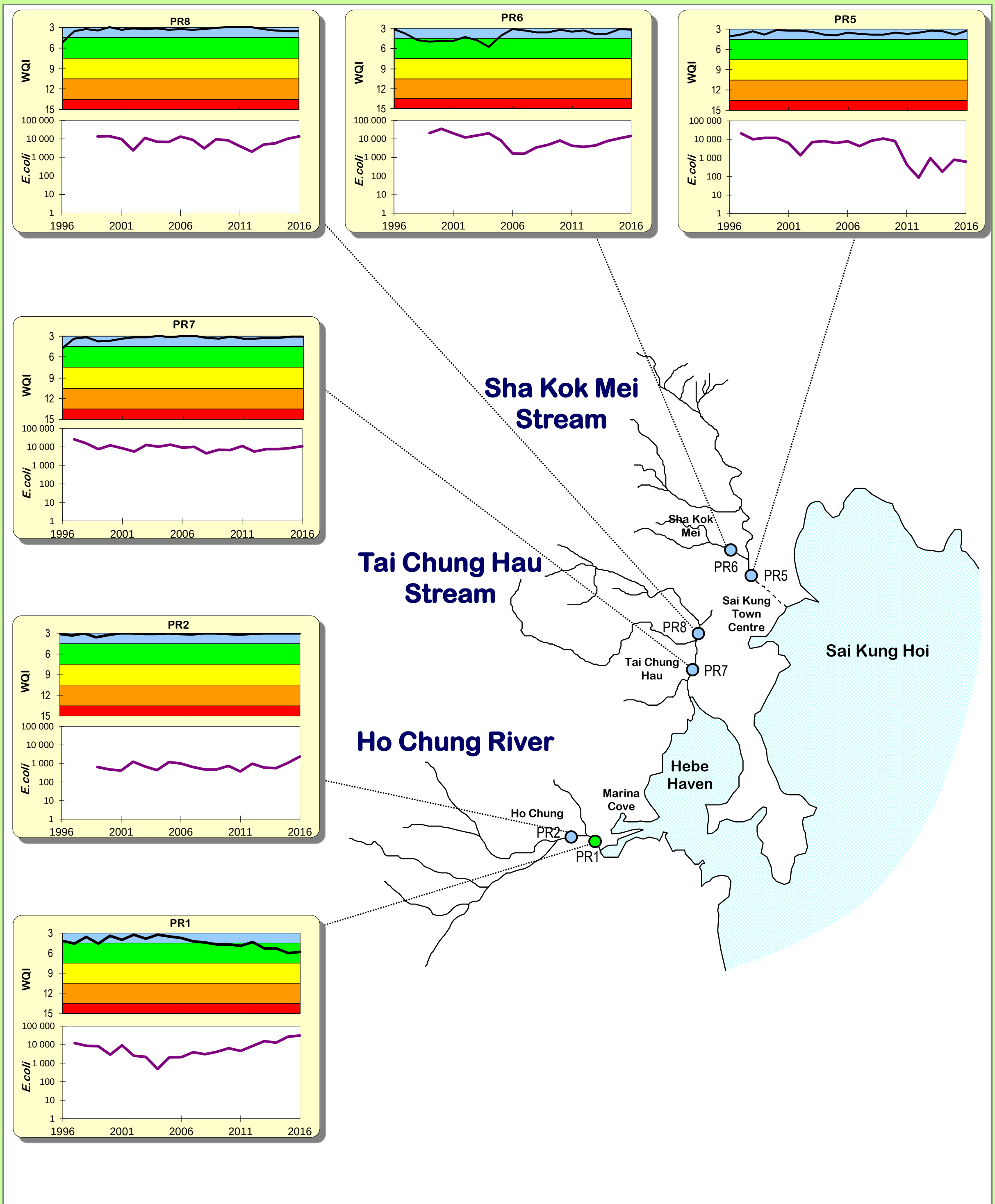
Water Quality Index gradings and *E. coli* levels in Tseng Lan Shue Stream



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

Water Quality Index gradings and *E. coli* levels in Ho Chung River, Tai Chung Hau Stream and Sha Kok Mei Stream



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

4. Northwestern New Territories rivers

In the Northwestern New Territories, the EPD monitors a total of 13 rivers and streams in the Deep Bay WCZ, which flow into Shenzhen River or directly into Deep Bay (Shenzhen Bay). These include River Indus, Beas and Ganges in the North District, Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah in the Yuen Long District, and the six smaller streams are located around the Lau Fau Shan area.

River Indus

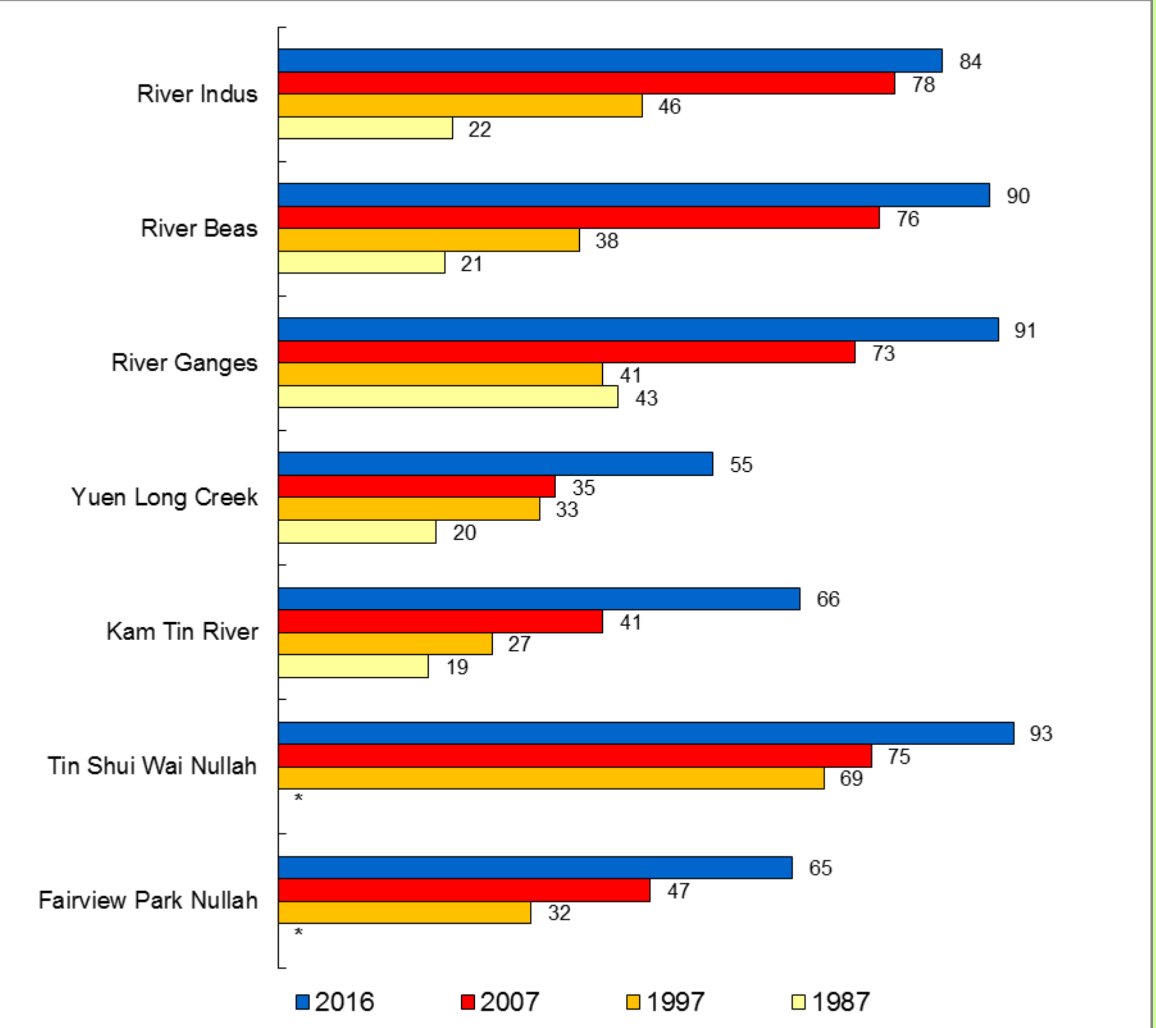
All the monitored rivers in the Northwestern New Territories showed improvement in the WQO compliance rates over the past three decades. In 2016, the overall WQO compliance rate was 83%, as compared with 70% in 2007, 55% in 1997 and 25% in 1987.

River Indus is a major river in the North District. It runs through rural areas like Lung Yeuk Tau, collects runoff from the densely populated Fanling and Sheung Shui urban areas, meets with River Beas before draining into Shenzhen River. The overall WQO compliance rate was 84% in 2016, as compared with 22% in 1987. In 2016, its downstream station (IN1), which is affected by backflow from Shenzhen River, had a “Fair” grading in 2016, same as that in 2015. The midstream (IN2) and upstream (IN3) stations were "Good" and "Excellent" respectively, similar to those in 2015.

As a tributary of River Indus, River Beas recorded an overall WQO compliance rate of 90% in 2016, as compared with 21% in 1987. Going downstream along River Beas, the three stations (RB1, RB2 and RB3) received "Excellent", "Good" and "Good" grading respectively in 2016, compared to "Good", "Good" and "Fair" in 2015.

For River Ganges, the overall WQO compliance rate was 91% in 2016. In terms of WQI, the downstream station GR1 was graded "Good", while the middle station GR2 and upstream station GR3 were graded "Excellent" in 2016.

Compliance with the Water Quality Objectives of the rivers in the Northwestern New Territories (North District and Yuen Long) over the past three decades in percentage



* Monitoring data for Tin Shui Wai Nullah and Fairview Park Nuallh in 1987 were not available

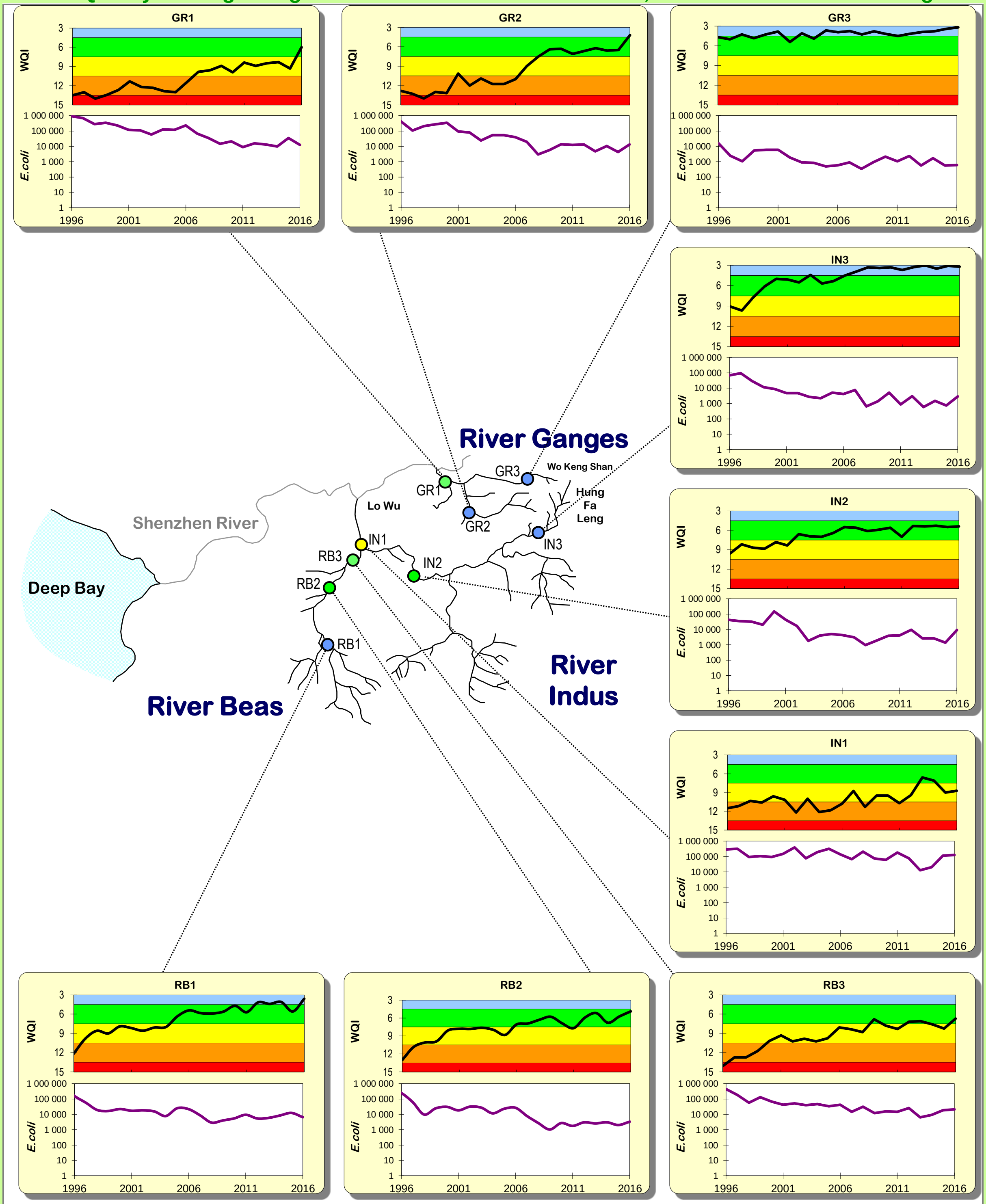
As a major river in Yuen Long District, Yuen Long Creek passes through rural areas as well as the densely populated Yuen Long new town and Yuen Long Kau Hui, joins Kam Tin River before entering Deep Bay. In 2016, Yuen Long Creek's overall compliance rate was 55%, compared with 35% in 2007 and 20% in 1987. For Kam Tin River, the overall compliance rate in 2016 was 66%. In terms of WQI, the two upstream stations (YL1 and YL2) recorded "Fair" and "Bad" grading, respectively, while the two downstream stations (YL3 and YL4) were graded "Bad" in 2016. Both stations in Kam Tin River (KT1 and KT2) also received WQI grading of "Bad" in 2016. Overall, this main river was still impacted by discharges from livestock farms, misconnections in old buildings and unsewered premises.

Yuen Long Creek passes through Yuen Long Town before flowing into Deep Bay

Tin Shui Wai Nullah reached an overall WQO compliance of 93% in 2016. In terms of WQI, its downstream station (TSR1) was graded "Fair" and its upstream station (TSR2) was graded "Excellent" in 2016.

The Fairview Park Nullah station (FVR1) recorded 65% WQO compliance in 2016 as compared with 47% in 2007 and 32% in 1997. In terms of WQI, it was graded "Fair" in 2016.

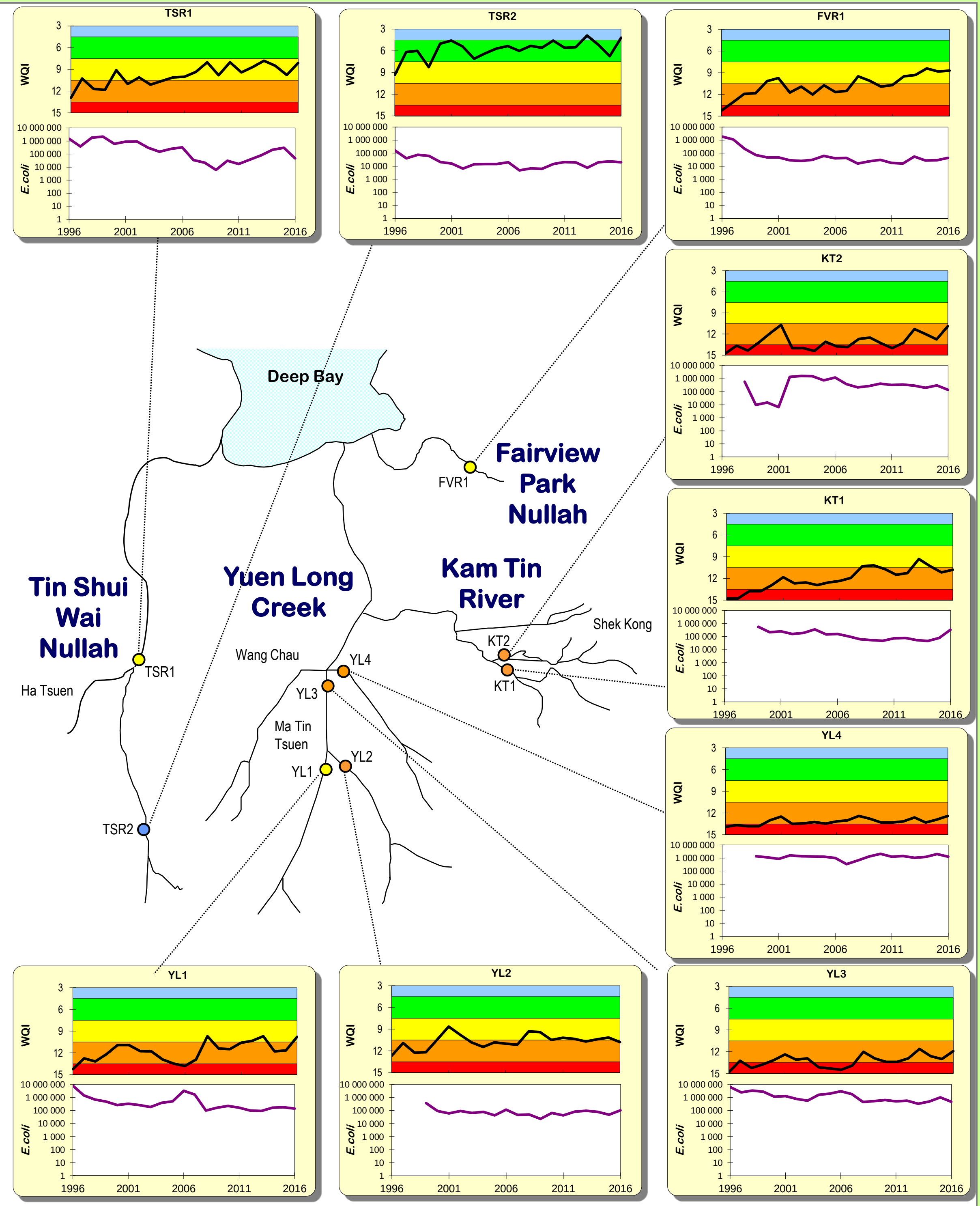
Water Quality Index gradings and *E. coli* levels in River Indus, River Beas and River Ganges



N.B.: WQI Annual Water Quality Index (Excellent, Good, Fair, Bad, Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

Water Quality Index gradings and *E. coli* levels in Yuen Long Creek, Kam Tin River, Tin Shui Wai Nullah and Fairview Park Nullah

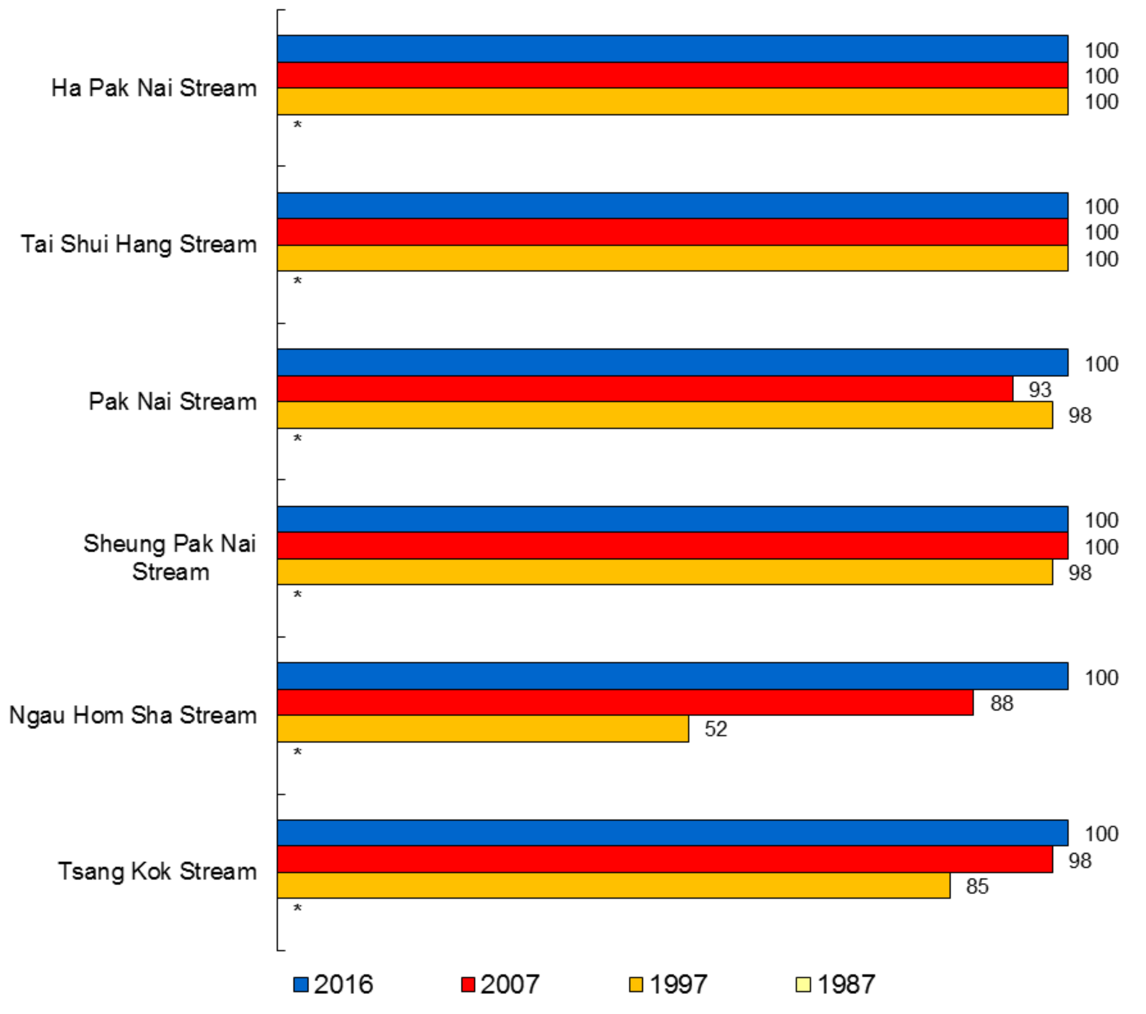


N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

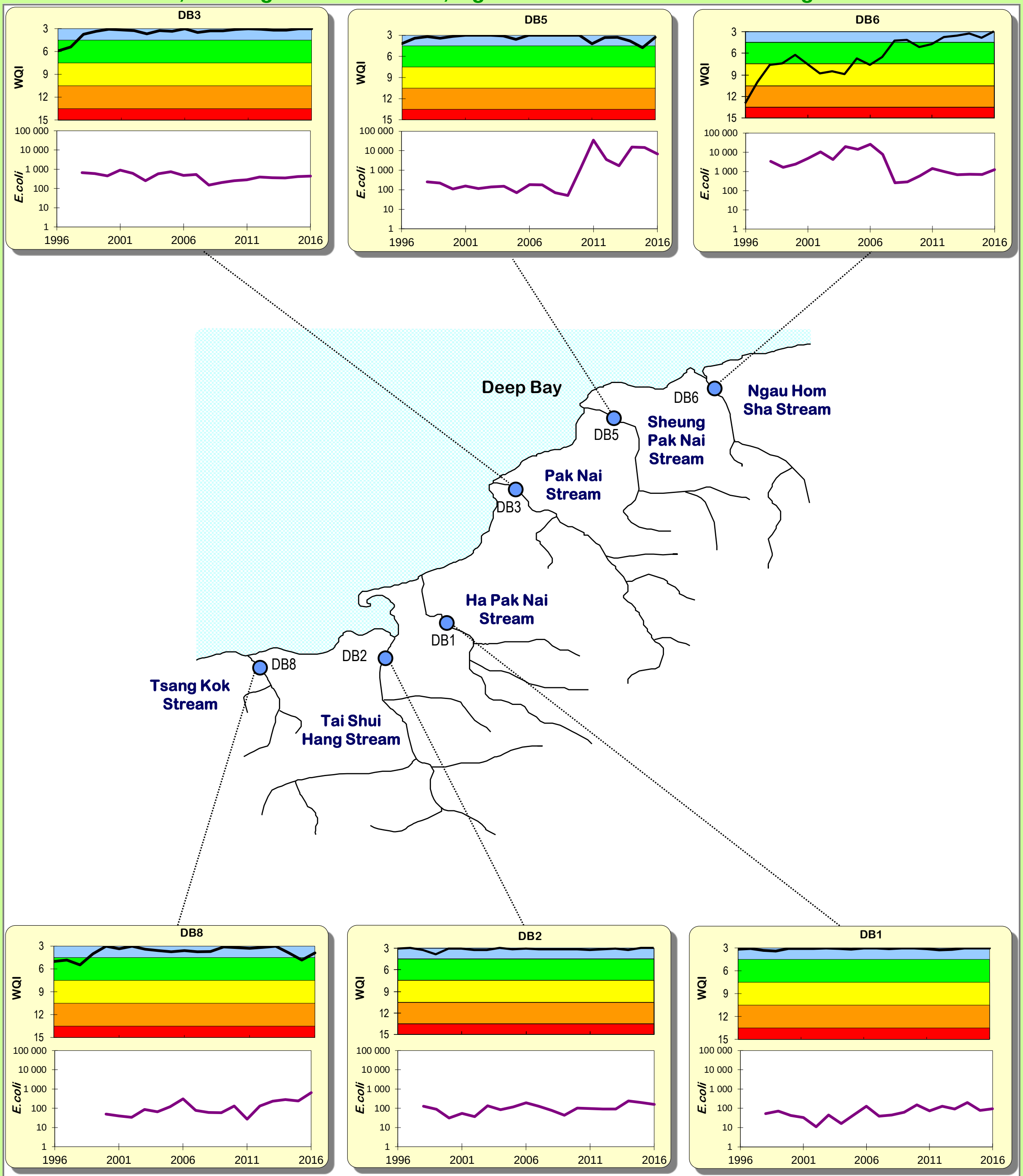
In 2016, the water quality of the six smaller streams around the Lau Fau Shan area remained good. All achieved an overall WQO compliance of 100% and “Excellent” WQI grading.

Compliance with the Water Quality Objectives of the of the smaller streams in Northwestern New Territories (Lau Fau Shan area) over the past two decades in percentage



* Monitoring data for all 6 smaller streams in Lau Fau Shan area in 1987 were not available

Water Quality Index gradings and *E. coli* levels in Ha Pak Nai Stream, Tai Shui Hang Stream, Pak Nai Stream, Sheung Pak Nai Stream, Ngau Hom Sha Stream and Tsang Kok Stream



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

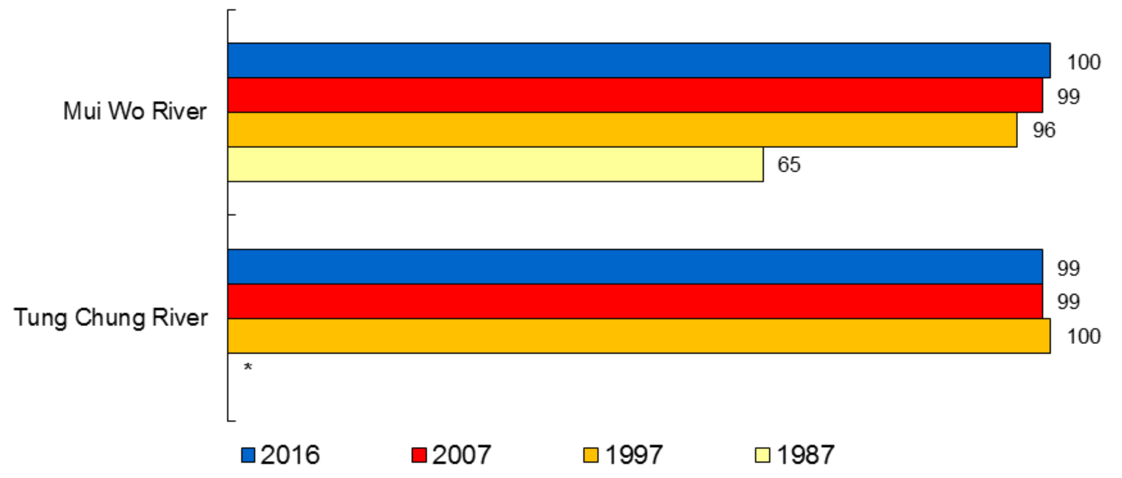
E. coli Annual geometric mean *E. coli* level (counts/100mL)

5. Lantau Island rivers

Lantau Island is large and less populated. Two rivers with a total of eight monitoring stations are monitored by the EPD: five along Mui Wo River on the south-eastern side of Lantau (Southern WCZ) and three at Tung Chung River on the north-western side of the island (North Western WCZ).

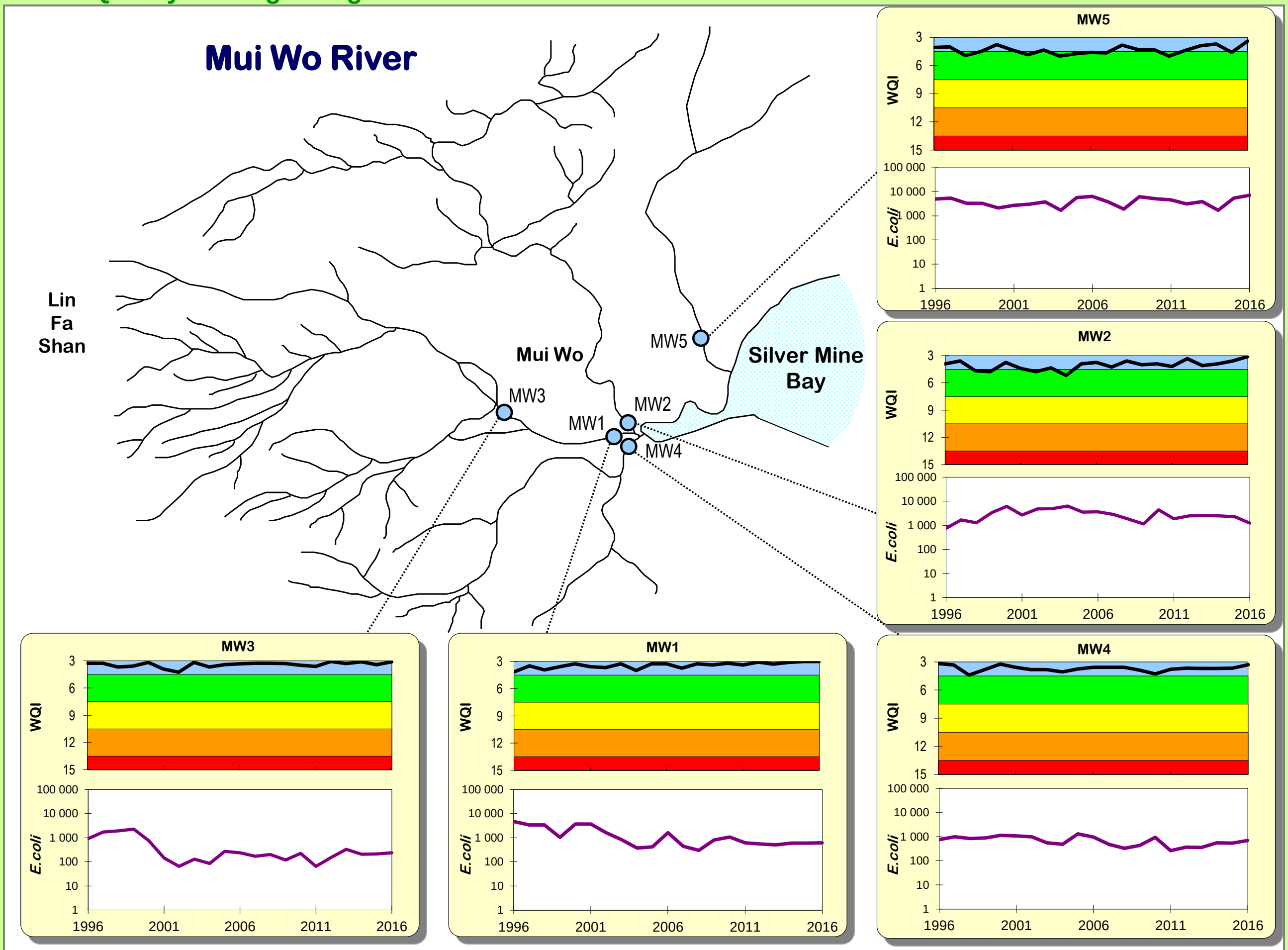
Overall, Mui Wo River and Tung Chung River generally displayed satisfactory water quality. The WQO compliance rates of Mui Wo River and Tung Chung River in 2016 were 100% and 99% respectively. As for the WQI grading, all five monitoring stations at Mui Wo River and three monitoring stations in Tung Chung River were graded “Excellent”.

Compliance with the Water Quality Objectives of the rivers in Lantau Island over the past three decades in percentage



* Monitoring data for Tung Chung River in 1987 were not available

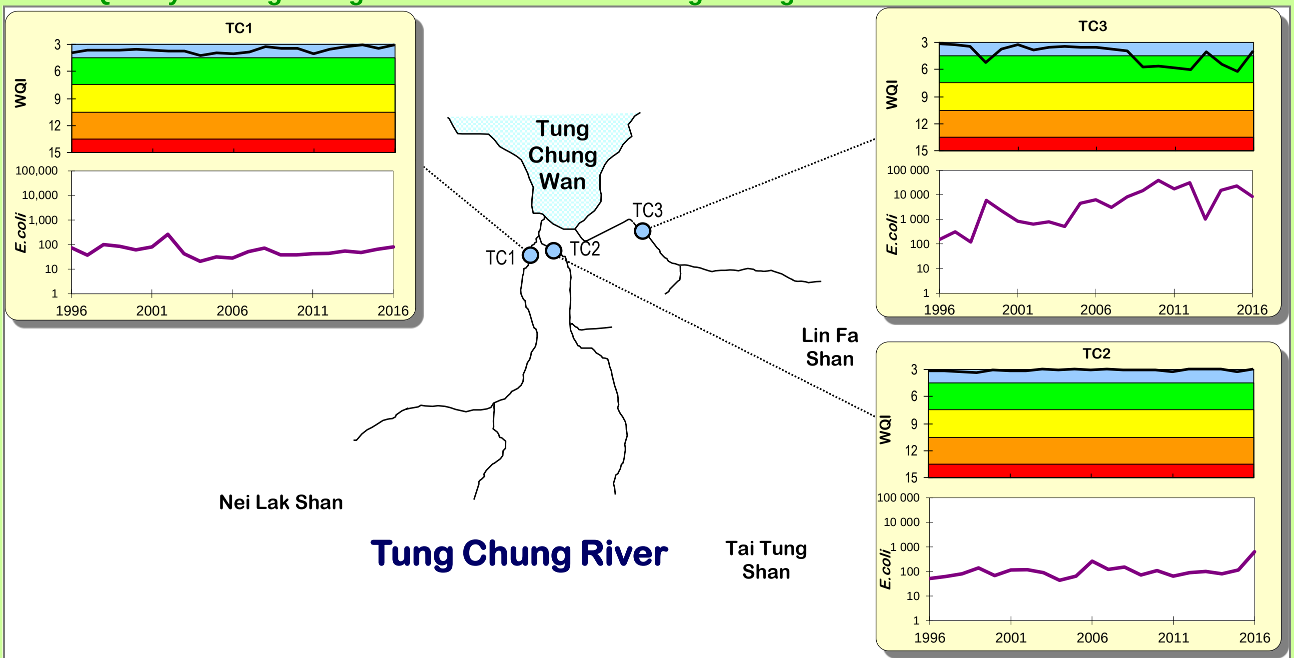
Water Quality Index gradings and *E. coli* levels in Mui Wo River



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

Water Quality Index gradings and *E. coli* levels in Tung Chung River



N.B.: WQI Annual Water Quality Index (Excellent Good Fair Bad Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

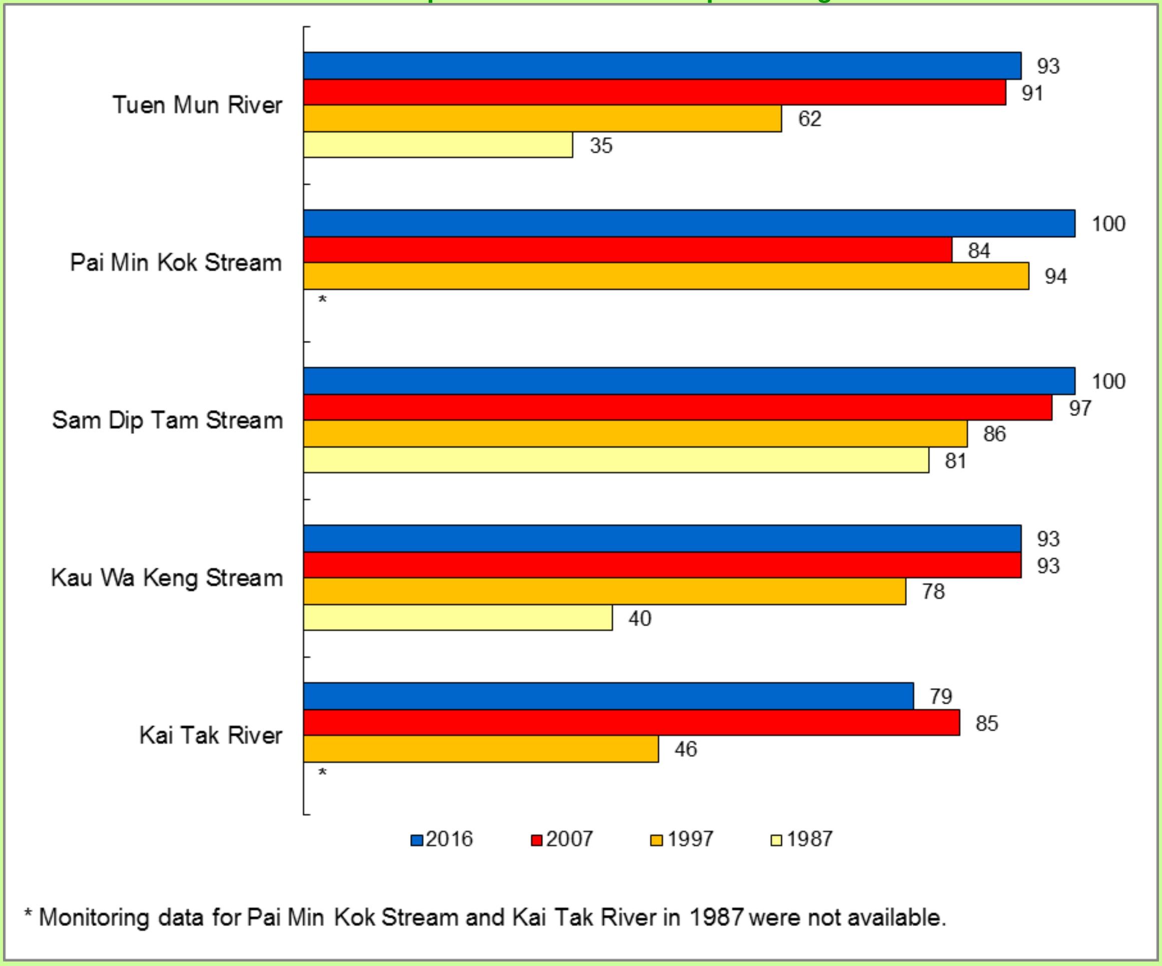
6. Southwestern New Territories and Kowloon rivers

The Southwestern New Territories and Kowloon area covers Tuen Mun to the eastern end of Victoria Harbour. The five watercourses being monitored are Tuen Mun River in Tuen Mun, Pai Min Kok Stream and Sam Dip Tam Stream near Tsuen Wan, Kau Wa Keng in Kwai Chung and Kai Tak River located in Kowloon. There was substantial improvement in the water quality of these urban watercourses over the past three decades.

Tuen Mun River

In 2016, 15 of the 18 stations (83%) were graded “Excellent” or “Good”, similar to 2015. Only one station was graded “Bad” this year (i.e. TN1 at Tuen Mun River), as in 2015 and 2014. The overall WQO compliance rate of the rivers and streams in the area has risen from 75% in 1997 to 93% in 2016.

Compliance with the Water Quality Objectives of the rivers in the Southwestern New Territories and Kowloon over the past three decades in percentage



As the major river in the Southwestern New Territories, Tuen Mun River passes through upstream areas like Lam Tei, San Hing Tsuen and Fu Tei, then the densely populated Tuen Mun town in its midstream, before entering Tuen Mun Typhoon Shelter. It showed marked improvement in the last three decades, with its WQO compliance rate rose steadily from 35% in 1987 to 93% in 2016. This improvement was attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun Sewage Master Plan. In 2016, the upstream station (TN1) was graded “Bad” mainly due to discharge from the unsewered areas. To prevent the pollutants from affecting Tuen Mun River, a dry weather flow interceptor (DWFI) has been installed at West Rail Siu Hong Station to divert the flow at TN1 to the foul sewers and sewage treatment works for treatment. All other five stations (TN2, TN3, TN4, TN5 and TN6) were graded “Good”.

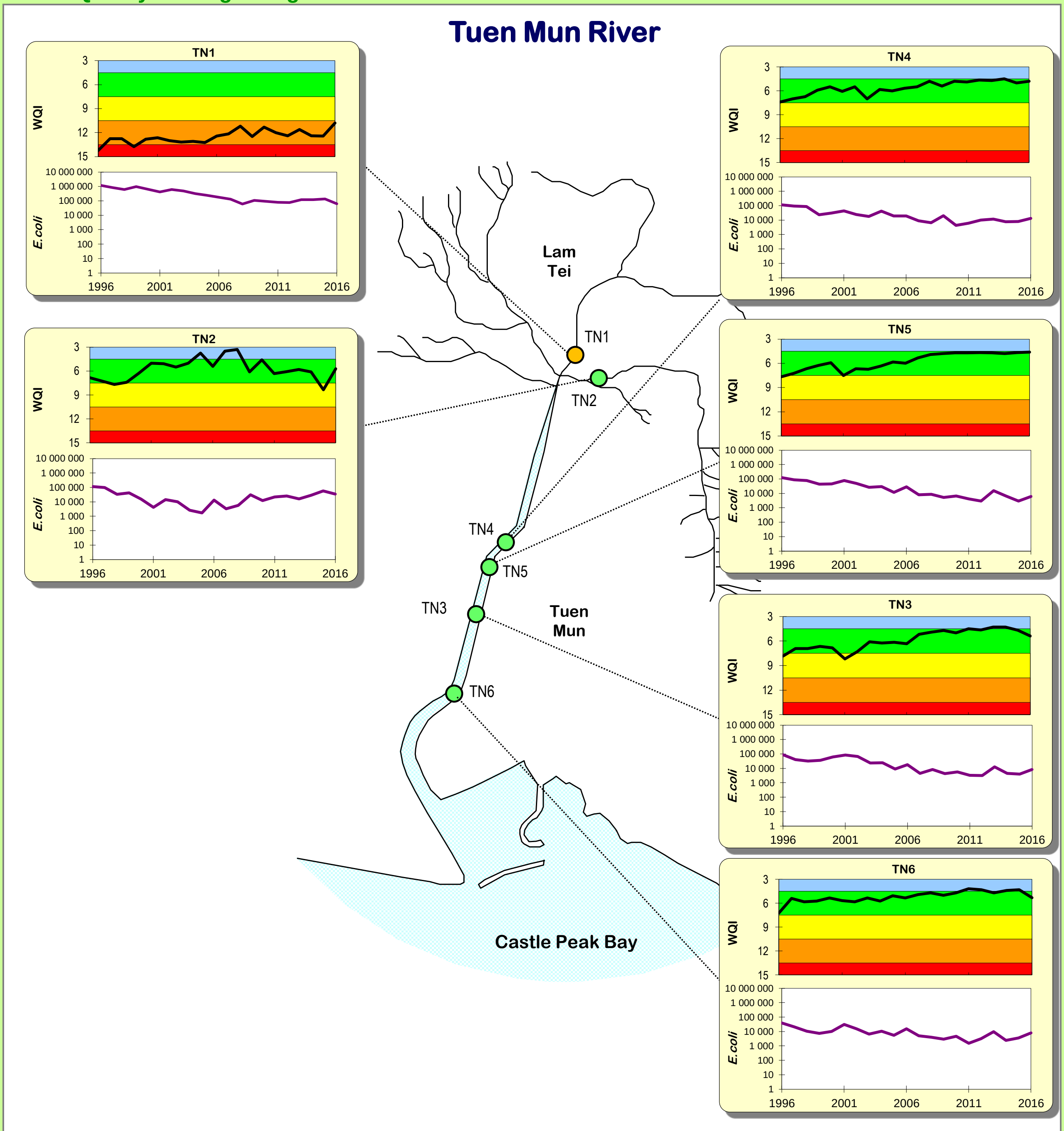
Pai Min Kok Stream achieved an overall WQO compliance rate of 100% in 2016. In terms of WQI, both its upstream station (AN2) and downstream station (AN1) were graded “Excellent” in 2016.

Sam Dip Tam Stream of Tsuen Wan also achieved 100% compliance with the WQO and all its three monitoring stations were graded “Excellent” in 2016.

The monitoring station KW3 at Kau Wa Keng Stream in Kwai Chung achieved a WQO compliance rate of 93% in 2016, as compared with 40% in 1987. Its WQI grading was “Good” in 2016.

Kai Tak River had a WQO compliance rate of 79% in 2016. The two upstream stations (KN5 and KN7) and two midstream stations (KN3 and KN4) were graded “Good”, while the two downstream stations (KN1 and KN2) were graded “Fair”. It was expected that the water quality of Kai Tak River would further improve once CEDD has completed its development work and DSD has repaired and upgraded the sewer system in its catchment area.

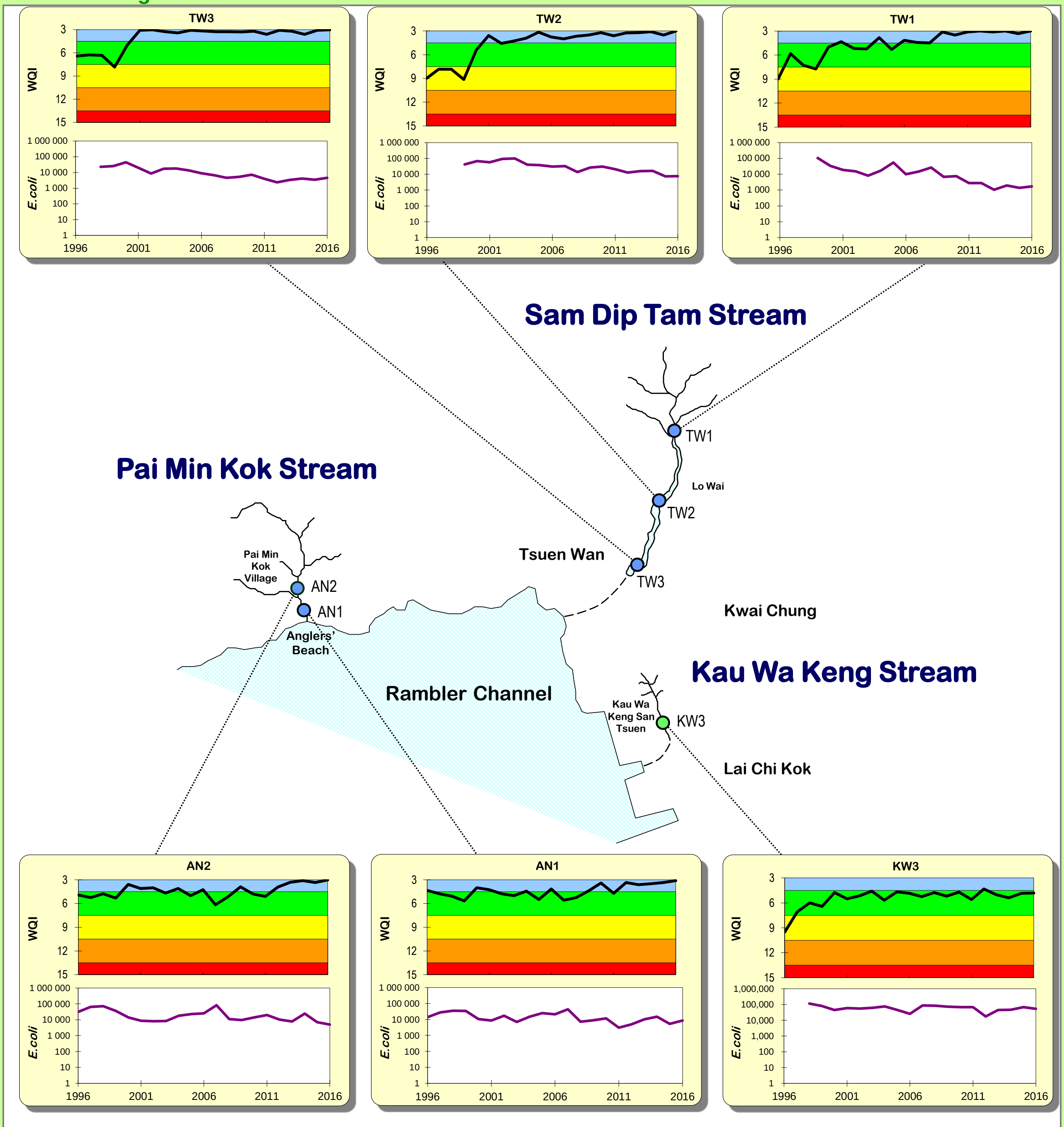
Water Quality Index gradings and *E. coli* levels in Tuen Mun River



N.B.: WQI — Annual Water Quality Index (Excellent, Good, Fair, Bad, Very Bad)

E. coli — Annual geometric mean *E. coli* level (counts/100mL)

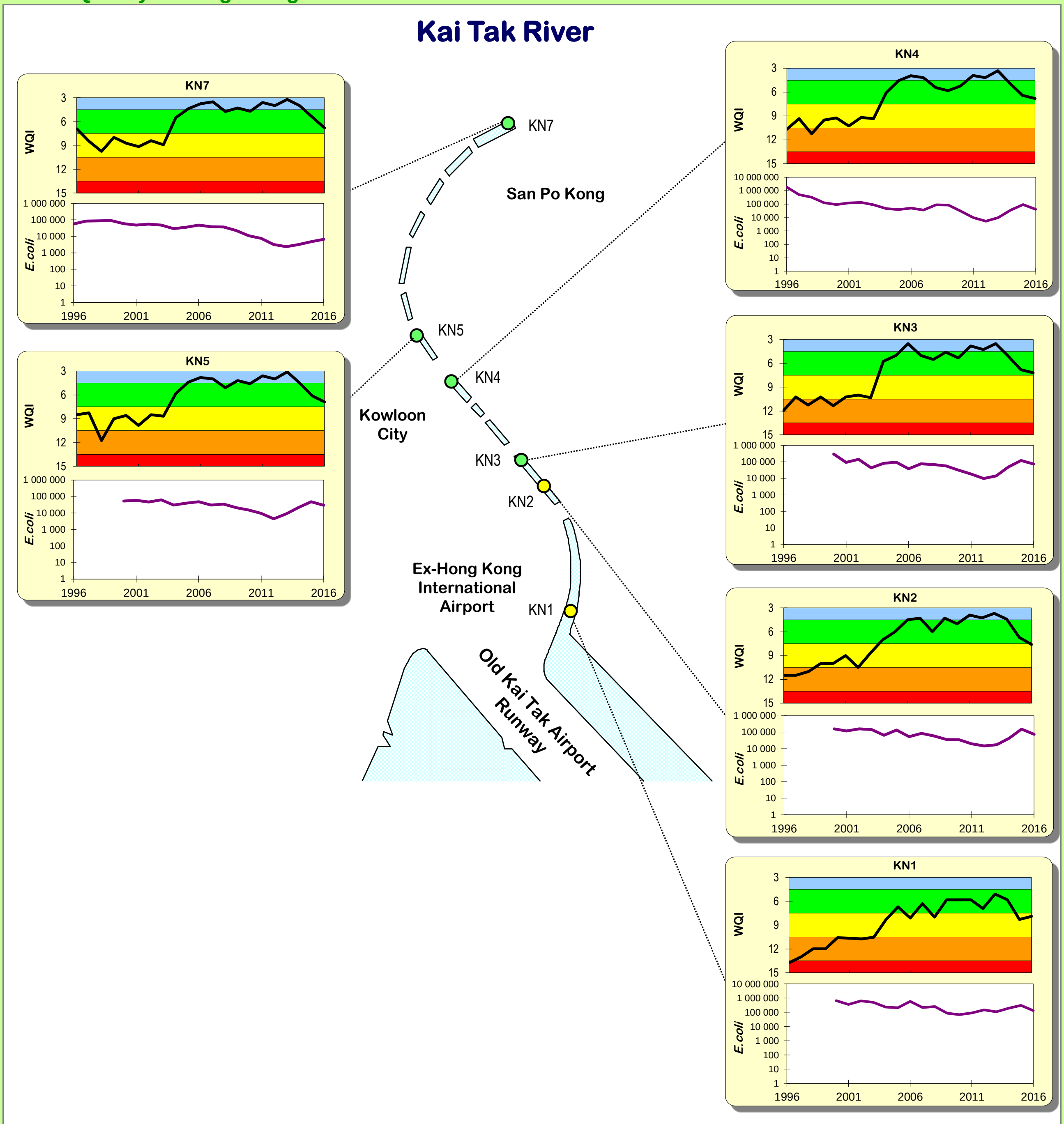
Water Quality Index gradings and *E. coli* levels in Pai Min Kok Stream, Sam Dip Tam Stream and Kau Wa Keng Stream



N.B.: WQI Annual Water Quality Index (Excellent, Good, Fair, Bad, Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

Water Quality Index gradings and *E. coli* levels in Kai Tak River



N.B.: WQI Annual Water Quality Index (Excellent, Good, Fair, Bad, Very Bad)

E. coli Annual geometric mean *E. coli* level (counts/100mL)

Appendices

Summary of river water quality monitoring stations and sampling frequencies in 2016

Area	Watercourse	Monitoring Station	(Number)	Sampling frequency
Eastern New Territories				
Sha Tin	Shing Mun River			
	Shing Mun Main Channel	TR19I	(1)	monthly
	Siu Lek Yuen Nullah	TR23A, TR23L	(2)	monthly
	Fo Tan Nullah	TR17, TR17L	(2)	monthly
	Kwun Yam Shan Stream	KY1	(1)	monthly
	Tai Wai Nullah	TR19, TR19A, TR19C	(3)	monthly
	Tin Sum Nullah	TR20B	(1)	monthly
Tai Po Town Centre	Lam Tsuen River	TR12, TR12B, TR12C, TR12D, TR12E, TR12F, TR12G, TR12H, TR12I	(9)	monthly
	Tai Po River	TR13	(1)	monthly
Tai Po Rural Area	Tai Po Kau Stream	TR14	(1)	monthly
	Shan Liu Stream	TR4	(1)	monthly
	Tung Tze Stream	TR6	(1)	monthly
Sai Kung	Ho Chung River	PR1, PR2	(2)	monthly
	Sha Kok Mei Stream	PR5, PR6	(2)	monthly
	Tai Chung Hau Stream	PR7, PR8	(2)	monthly
Tseung Kwan O	Tseng Lan Shue Stream	JR3, JR6, JR11	(3)	monthly
Northwestern New Territories				
North District	River Indus (Ng Tung River)	IN1, IN2, IN3	(3)	monthly
	River Beas (Sheung Yue River)	RB1, RB2, RB3	(3)	monthly
	River Ganges (Ping Yuen River)	GR1, GR2, GR3	(3)	monthly
Yuen Long	Yuen Long Creek	YL1, YL2, YL3, YL4	(4)	monthly
	Kam Tin River	KT1, KT2	(2)	monthly
	Tin Shui Wai Nullah	TSR1, TSR2	(2)	monthly
	Fairview Park Nullah	FVR1	(1)	monthly
Lau Fau Shan	Ha Pak Nai Stream	DB1	(1)	monthly
	Tai Shui Hang Stream	DB2	(1)	monthly
	Pak Nai Stream	DB3	(1)	monthly
	Sheung Pak Nai Stream	DB5	(1)	monthly
	Ngau Hom Sha Stream	DB6	(1)	monthly
	Tsang Kok Stream	DB8	(1)	monthly
Lantau Island				
Mui Wo	Mui Wo River	MW1, MW2, MW3, MW4, MW5	(5)	monthly
Tung Chung	Tung Chung River	TC1, TC2, TC3	(3)	monthly
Southwestern New Territories and Kowloon				
Tuen Mun	Tuen Mun River	TN1, TN2, TN3, TN4, TN5, TN6	(6)	monthly
Tsuen Wan and Kwai Chung	Pai Min Kok (Anglers') Stream	AN1, AN2	(2)	monthly
	Sam Dip Tam Stream	TW1, TW2, TW3	(3)	monthly
	Kau Wa Keng Stream	KW3	(1)	monthly
Kowloon	Kai Tak River	KN1, KN2, KN3, KN4, KN5, KN7	(6)	monthly
Total	**30**	**82**		**-**

River water quality parameters and the methods of analysis (part 1 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method[1] / Analyst
Physical Chemical Properties		
Water temperature	0.1 °C	Multi-parameter water quality data logger, model YSI-6820 / On-site measurement / EPD
Dissolved Oxygen	0.1 mg/L, 1%	
pH	0.1	
Conductivity	1 μS/cm	
Turbidity	0.1 NTU	
Flow	1 L/s	Electromagnetic flow meter, model Flo-mate 2000 / On-site measurement / EPD
Solid Contents		
Suspended Solids	0.5 mg/L	In-house method GL-PH-23, based on APHA[2] 22ed 2540 D / Government Laboratory
Total Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 22ed 2540 F / Government Laboratory
Total Volatile Solids	0.5 mg/L	In-house method GL-PH-19, based on APHA 20ed 2540 E / Government Laboratory
Aggregate Organics		
5-Day Biochemical Oxygen Demand (BOD_5)	1 mg/L	In-house method based on APHA 18ed 5210 B / EPD
Chemical Oxygen Demand (COD)	2 mg/L	In-house method GL-OR-38 & GL-OR-39, based on ASTM[3] D1252-06, Method B or APHA 22ed 5220 C & D / Government Laboratory
Total Organic Carbon (TOC)	1 mg/L	In-house method GL-OR-32, based on APHA 20ed 5310 B / Government Laboratory
Faecal Bacteria		
E. coli	1 counts /100 mL	In-house method[4], membrane filtration with CHROMagar Liquid ECC medium / EPD
Faecal Coliforms	1 counts / 100 mL	
Nutrients		
Ammonia-Nitrogen	0.005 mg/L	In-house method GL-IN-15, based on APHA 22ed 4500-NH_3 / Government Laboratory
Nitrite-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO_2^- B (FIA) / Government Laboratory
Nitrate-Nitrogen	0.002 mg/L	In-house method GL-IN-18, based on APHA 22ed 4500-NO_3^- F & I (FIA) / Government Laboratory
Total Kjeldahl Nitrogen	0.05 mg/L	In-house method GL-IN-14 & GL-IN-15, based on ASTM D3590-11 B (FIA) & APHA 20ed 4500-N A&D (FIA) / Government Laboratory
Ortho-phosphate Phosphorus	0.002 mg/L	In-house method GL-IN-16, based on ASTM D515-88 A (FIA) / Government Laboratory
Total Phosphorus	0.02 mg/L	In-house method GL-IN-14 & GL-IN-16, based on ASTM D515-88 B (FIA) & APHA 20ed 4500-P G (FIA) / Government Laboratory
Molybdate-Reactive Silica	0.05 mg/L	In-house method GL-IN-17, based on APHA 20ed 4500-SiO2 C&E (FIA) / Government Laboratory

Reference notes:

1. *Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.*
2. *APHA - American Public Health Association: Standard Methods for the Examination of Water and Wastewater.*
3. *ASTM - Annual Book of American Society for the Testing and Materials Standards, Vol. 11.01 & 11.02.*
4. i) Ho, B.S.W. and Tam, T.Y. (1997). Enumeration of *E. coli* in environmental waters and wastewater using a chromogenic medium. *Wat. Sci. Tech.*, **35**, 409-413.
   ii) DoE and DHSS (1983). "The bacteriological examination of drinking water supplies 1982. Report on Public Health and Medical Subjects No. 71. Methods for the Examination of Waters and Associated Materials". Department of Environment, Department of Health and Social Security, Public Health Laboratory Service, H.M.S.O. London.

River water quality parameters and the methods of analysis (part 2 of 2)

Water Quality Parameter	Reporting Limit and Unit	Analytical Method[1] / Analyst
Metals		
Aluminium	50 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Antimony	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Arsenic	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Barium	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Beryllium	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Boron	50 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Cadmium	0.1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Chromium	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Copper	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Iron	50 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Lead	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Manganese	10 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Mercury	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Molybdenum	2 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Nickel	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Silver	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Thallium	1 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Vanadium	2 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Zinc	10 μg/L	In-house method GL-TE-63, based on APHA 22ed 3111, 3112, 3113, 3114 & 3120 / Government Laboratory
Industrial and Commercial Pollutants		
Cyanide	0.01 mg/L	In-house method GL-IN-42, based on ASTM D 2036-09 or APHA 22ed 4500-CN / Government Laboratory
Fluoride	0.2 mg/L	In-house method GL-IN-47, based on APHA 22ed 4500-F^- C & G (Ion Selective Electrode) and ASTM D1179-99 B (FIA) / Government Laboratory
Anionic Surfactants (as Manoxol OT)	0.05 mg/L	In-house method GL-OR-30, based on BS 6068, Section 2.23 (1994) (Colorimetric) & In-house method GL-OR-65, based on Abbott, D.C. “Analyst”, Vol.87, p.286(1962) & S. Motomizu et al., “Analyst” Vol.113, p.747(1988) (FIA) / Government Laboratory
Oil and Grease	0.5 mg/L	In-house method GL-OR-26, based on APHA 22ed 5520 B / Government Laboratory
Sulphide Contents		
Free Hydrogen Sulphide	0.01 mg/L	In-house method GL-IN-46, based on APHA 22ed 4500S^{2-} D (Colorimetric) / Government Laboratory
Sulphide	0.02 mg/L	In-house method GL-IN-46, based on APHA 22ed 4500S^{2-} D (Colorimetric) / Government Laboratory
Plant Pigments		
Chlorophyll-*a*	0.2 μg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 (spectrophotometric) / Government Laboratory
Pheo-Pigment	0.2 μg/L	In house method GL-OR-34, based on APHA 20ed 10200H 2 (spectrophotometric) / Government Laboratory

Reference notes:

1. *Mention of brand names of commercial products does not constitute or imply endorsement or recommendation by the Environmental Protection Department.*
2. *APHA - American Public Health Association: Standard Methods for the Examination of Water and Wastewater.*
3. *ASTM - Annual Book of American Society for the Testing and Materials Standards, Vol. 11.01 & 11.02.*

Key Water Quality Objectives (WQOs) for river monitoring stations in the Eastern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Tolo Harbour and Channel Water Control Zone						
Shing Mun River	KY1	6.5 - 8.5	3	15	20	4
	TR17	6.5 - 8.5	5	30	20	4
	TR17L	6.5 - 8.5	5	30	20	4
	TR19	6.5 - 8.5	5	30	20	4
	TR19A	6.5 - 8.5	5	30	20	4
	TR19C	6.5 - 8.5	5	30	20	4
	TR19I	6.0 - 9.0	5	30	25	4
	TR20B	6.5 - 8.5	5	30	20	4
	TR23A	6.5 - 8.5	3	15	20	4
	TR23L	6.5 - 8.5	3	15	20	4
Lam Tsuen River	TR12	6.5 - 8.5	3	15	20	4
	TR12B	6.5 - 8.5	3	15	20	4
	TR12C	6.5 - 8.5	3	15	20	4
	TR12D	6.5 - 8.5	3	15	20	4
	TR12E	6.5 - 8.5	3	15	20	4
	TR12F	6.5 - 8.5	3	15	20	4
	TR12G	6.5 - 8.5	3	15	20	4
	TR12H	6.5 - 8.5	3	15	20	4
	TR12I	6.0 - 9.0	5	30	25	4
Tai Po River	TR13	6.5 - 8.5	5	30	20	4
Tai Po Kau Stream	TR14	6.0 - 9.0	5	30	25	4
Shan Liu Stream	TR4	6.0 - 9.0	5	30	25	4
Tung Tze Stream	TR6	6.0 - 9.0	5	30	25	4
Port Shelter Water Control Zone						
Ho Chung River	PR1	6.5 - 8.5	5	30	25	4
	PR2	6.5 - 8.5	5	30	25	4
Sha Kok Mei Stream	PR5	6.0 - 9.0	5	30	25	4
	PR6	6.0 - 9.0	5	30	25	4
Tai Chung Hau Stream	PR7	6.0 - 9.0	5	30	25	4
	PR8	6.0 - 9.0	5	30	25	4
Junk Bay Water Control Zone						
Tseng Lan Shue Stream	JR3	6.0 - 9.0	5	30	25	4
	JR6	6.0 - 9.0	5	30	25	4
	JR11	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Northwestern New Territories

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxygen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Deep Bay Water Control Zone						
River Indus	IN1	6.5 - 8.5	3	15	20	4
	IN2	6.5 - 8.5	3	15	20	4
	IN3	6.5 - 8.5	3	15	20	4
River Beas	RB1	6.5 - 8.5	3	15	20	4
	RB2	6.5 - 8.5	3	15	20	4
	RB3	6.5 - 8.5	3	15	20	4
River Ganges	GR1	6.5 - 8.5	3	15	20	4
	GR2	6.5 - 8.5	3	15	20	4
	GR3	6.5 - 8.5	3	15	20	4
Yuen Long Creek	YL1	6.5 - 8.5	3	15	20	4
	YL2	6.5 - 8.5	3	15	20	4
	YL3	6.5 - 8.5	5	30	20	4
	YL4	6.5 - 8.5	5	30	20	4
Kam Tin River	KT1	6.5 - 8.5	3	15	20	4
	KT2	6.5 - 8.5	3	15	20	4
Tin Shui Wai Nullah	TSR1	6.0 - 9.0	5	30	20	4
	TSR2	6.0 - 9.0	5	30	20	4
Fairview Park Nullah	FVR1	6.0 - 9.0	5	30	20	4
Ha Pak Nai Stream	DB1	6.0 - 9.0	5	30	20	4
Tai Shui Hang Stream	DB2	6.0 - 9.0	5	30	20	4
Pak Nai Stream	DB3	6.0 - 9.0	5	30	20	4
Sheung Pak Nai Stream	DB5	6.0 - 9.0	5	30	20	4
Ngau Hom Sha Stream	DB6	6.0 - 9.0	5	30	20	4
Tsang Kok Stream	DB8	6.0 - 9.0	5	30	20	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations on Lantau Island

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxgyen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
Southern Water Control Zone						
Mui Wo River	MW1	6.5 - 8.5	5	30	20	4
	MW2	6.5 - 8.5	5	30	20	4
	MW3	6.5 - 8.5	5	30	20	4
	MW4	6.5 - 8.5	5	30	20	4
	MW5	6.0 - 9.0	5	30	25	4
North Western Water Control Zone						
Tung Chung River	TC1	6.0 - 9.0	5	30	25	4
	TC2	6.0 - 9.0	5	30	25	4
	TC3	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Key Water Quality Objectives (WQOs) for river monitoring stations in the Southwestern New Territories and Kowloon

Watercourse	Monitoring station	Key Water Quality Objective				
		pH range	Maximum 5-Day Biochemical Oxgyen Demand (mg/L)	Maximum Chemical Oxygen Demand (mg/L)	Maximum Annual Median Suspended Solids (mg/L)	Minimum Dissolved Oxygen (mg/L)
North Western Water Control Zone						
Tuen Mun River	TN1	6.0 - 9.0	5	30	25	4
	TN2	6.5 - 8.5	3	15	20	4
	TN3	6.0 - 9.0	5	30	25	4
	TN4	6.0 - 9.0	5	30	25	4
	TN5	6.0 - 9.0	5	30	25	4
	TN6	6.0 - 9.0	5	30	25	4
Western Buffer Water Control Zone						
Pai Min Kok (Anglers') Stream	AN1	6.0 - 9.0	5	30	25	4
	AN2	6.0 - 9.0	5	30	25	4
Victoria Harbour Water Control Zone						
Sam Dip Tam Stream	TW1	6.0 - 9.0	5	30	25	4
	TW2	6.0 - 9.0	5	30	25	4
	TW3	6.0 - 9.0	5	30	25	4
Kau Wa Keng Stream	KW3	6.0 - 9.0	5	30	25	4
Kai Tak River	KN1	6.0 - 9.0	5	30	25	4
	KN2	6.0 - 9.0	5	30	25	4
	KN3	6.0 - 9.0	5	30	25	4
	KN4	6.0 - 9.0	5	30	25	4
	KN5	6.0 - 9.0	5	30	25	4
	KN7	6.0 - 9.0	5	30	25	4

* The WQO compliance for suspended solids is based on annual median value, while WQO compliance for other parameters is based on individual measurements.

Summary of water quality monitoring data for Shing Mun River (Main Channel and Siu Lek Yuen Nullah) in 2016

Parameter	Unit	Shing Mun Main Channel	Siu Lek Yuen Nullah	
		TR19I	TR23L	TR23A
Dissolved oxygen	mg/L	7.2 (5.4 - 9.0)	8.7 (7.8 - 10.1)	7.1 (5.7 - 9.3)
pH		7.7 (7.2 - 8.2)	8.6 (7.3 - 8.9)	7.4 (7.2 - 8.1)
Suspended solids	mg/L	4 (2 - 59)	1 (<1 - 80)	4 (<1 - 39)
5-day Biochemical Oxygen Demand	mg/L	3 (<1 - 4)	<1 (<1 - 2)	2 (<1 - 8)
Chemical Oxygen Demand	mg/L	12 (7 - 18)	3 (<2 - 16)	8 (6 - 19)
Oil & grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.8)
E. coli	counts/ 100mL	3 900 (330 - 67 000)	1 400 (110 - 6 200)	16 000 (600 - 140 000)
Faecal coliforms	counts/ 100mL	35 000 (2 200 - 310 000)	18 000 (760 - 85 000)	62 000 (1 300 - 320 000)
Ammonia-nitrogen	mg/L	0.16 (0.09 - 0.32)	0.01 (<0.01 - 0.03)	0.36 (0.11 - 2.50)
Nitrate-nitrogen	mg/L	0.34 (0.04 - 0.78)	0.21 (0.16 - 0.35)	0.37 (0.25 - 0.54)
Total Kjeldahl Nitrogen	mg/L	0.42 (0.23 - 0.83)	0.13 (<0.05 - 0.37)	0.67 (0.19 - 2.90)
Ortho-phosphate Phosphorus	mg/L	<0.01 (<0.01 - 0.04)	<0.01 (<0.01 - 0.13)	0.02 (<0.01 - 0.16)
Total phosphorus	mg/L	0.07 (0.03 - 0.11)	<0.02 (<0.02 - 0.15)	0.07 (0.04 - 0.22)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 259)	88 (<50 - 492)	92 (<50 - 821)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 1)	<1 (<1 - 1)
Copper	µg/L	4 (2 - 6)	<1 (<1 - 2)	2 (2 - 9)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - 5)	<1 (<1 - 13)
Zinc	µg/L	19 (10 - 32)	15 (<10 - 24)	20 (13 - 96)
Flow	L/s	NM	39 (14 - 1 968)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Fo Tan Nullah and Kwun Yam Shan Stream) in 2016

Parameter	Unit	Fo Tan Nullah		Kwun Yam Shan Stream
		TR17	TR17L	KY1
Dissolved oxygen	mg/L	8.7 (7.1 - 10.6)	7.3 (5.9 - 8.5)	8.5 (7.8 - 9.7)
pH		8.0 (7.3 - 9.4)	7.5 (7.2 - 8.7)	8.5 (7.5 - 9.0)
Suspended solids	mg/L	6 (2 - 12)	4 (2 - 12)	5 (2 - 28)
5-day Biochemical Oxygen Demand	mg/L	3 (1 - 30)	2 (<1 - 6)	<1 (<1 - 1)
Chemical Oxygen Demand	mg/L	7 (3 - 22)	11 (4 - 22)	3 (<2 - 6)
Oil & grease	mg/L	<0.5 (<0.5 - 2.1)	<0.5 (<0.5 - 1.3)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	4 200 (100 - 80 000)	11 000 (900 - 58 000)	570 (68 - 3 900)
Faecal coliforms	counts/ 100mL	22 000 (2 000 - 260 000)	98 000 (12 000 - 490 000)	4 300 (670 - 20 000)
Ammonia-nitrogen	mg/L	0.03 (0.02 - 0.17)	0.20 (0.04 - 0.58)	0.02 (<0.01 - 0.04)
Nitrate-nitrogen	mg/L	0.69 (0.22 - 1.20)	0.56 (0.26 - 0.94)	0.82 (0.54 - 1.20)
Total Kjeldahl nitrogen	mg/L	0.62 (0.18 - 1.40)	0.54 (0.32 - 0.96)	0.16 (0.06 - 0.30)
Ortho-phosphate Phosphorus	mg/L	<0.01 (<0.01 - 0.07)	0.02 (<0.01 - 0.08)	0.07 (0.02 - 0.12)
Total phosphorus	mg/L	0.09 (0.02 - 0.23)	0.08 (0.06 - 0.11)	0.09 (0.07 - 0.15)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	197 (64 - 685)	136 (<50 - 400)	78 (59 - 170)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 11)	<1 (<1 - 3)	<1 (<1 - <1)
Copper	µg/L	3 (2 - 8)	4 (2 - 5)	1 (<1 - 3)
Lead	µg/L	<1 (<1 - 17)	1 (<1 - 9)	<1 (<1 - 3)
Zinc	µg/L	32 (19 - 46)	27 (15 - 44)	14 (<10 - 27)
Flow	L/s	80 (43 - 890)	NM	8 (5 - 28)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Shing Mun River (Tai Wai Nullah and Tin Sum Nullah) in 2016

Parameter	Unit	Tai Wai Nullah			Tin Sum Nullah
		TR19A	TR19C	TR19	TR20B
Dissolved oxygen	mg/L	8.8 (8.1 - 10.4)	8.9 (8.0 - 11.0)	9.2 (8.1 - 11.9)	8.5 (7.9 - 9.5)
pH		8.2 (7.4 - 9.5)	7.8 (7.3 - 9.0)	7.8 (7.3 - 8.7)	8.5 (7.5 - 11.4)
Suspended solids	mg/L	4 (1 - 14)	3 (2 - 140)	3 (1 - 11)	4 (<1 - 77)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 3)	<1 (<1 - 4)	<1 (<1 - 3)	<1 (<1 - 3)
Chemical Oxygen Demand	mg/L	5 (2 - 11)	4 (2 - 13)	4 (<2 - 15)	4 (<2 - 12)
Oil & grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.8)
E. coli	counts/ 100mL	3 700 (150 - 42 000)	6 500 (980 - 51 000)	7 100 (800 - 45 000)	44 (<1 - 51 000)
Faecal coliforms	counts/ 100mL	25 000 (2 700 - 220 000)	29 000 (4 000 - 270 000)	43 000 (9 000 - 290 000)	160 (<1 - 59 000)
Ammonia-nitrogen	mg/L	0.03 (0.01 - 0.06)	0.04 (0.02 - 0.11)	0.05 (0.01 - 0.13)	0.05 (<0.01 - 0.12)
Nitrate-nitrogen	mg/L	0.96 (0.28 - 1.30)	0.88 (0.26 - 1.00)	0.91 (0.31 - 1.00)	0.91 (0.51 - 1.50)
Total Kjeldahl nitrogen	mg/L	0.20 (0.05 - 0.54)	0.31 (0.14 - 0.65)	0.35 (0.11 - 0.97)	0.26 (0.11 - 3.80)
Ortho-phosphate Phosphorus	mg/L	<0.01 (<0.01 - 0.02)	0.02 (<0.01 - 0.04)	0.02 (<0.01 - 0.04)	0.01 (<0.01 - 0.15)
Total phosphorus	mg/L	0.03 (<0.02 - 0.08)	0.04 (<0.02 - 0.12)	0.04 (0.02 - 0.16)	0.03 (<0.02 - 0.67)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	100 (58 - 267)	98 (69 - 433)	96 (60 - 413)	261 (92 - 735)
Cadmium	µg/L	0.4 (0.1 - 0.8)	0.1 (<0.1 - 0.2)	0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)	3 (<1 - 27)
Copper	µg/L	1 (<1 - 4)	3 (2 - 7)	4 (2 - 7)	2 (1 - 13)
Lead	µg/L	<1 (<1 - 5)	1 (<1 - 10)	<1 (<1 - 12)	<1 (<1 - 5)
Zinc	µg/L	43 (29 - 61)	35 (22 - 88)	31 (22 - 81)	20 (10 - 77)
Flow	L/s	29 (9 - 459)	84 (23 - 1 185)	78 (66 - 2 490)	64 (18 - 109)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2016 (Part 1 of 3)

Parameter	Unit	Lam Tsuen River		
		TR12H	TR12D	TR12C
Dissolved oxygen	mg/L	8.6 (7.8 - 10.2)	8.8 (7.8 - 10.3)	8.4 (7.5 - 9.6)
pH		7.4 (7.3 - 7.6)	7.4 (7.1 - 7.7)	7.6 (7.4 - 7.8)
Suspended solids	mg/L	1 (<1 - 17)	<1 (<1 - 4)	3 (1 - 10)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	<1 (<1 - <1)	2 (1 - 8)
Chemical Oxygen Demand	mg/L	3 (<2 - 5)	<2 (<2 - 8)	7 (4 - 14)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	1 500 (510 - 7 500)	370 (80 - 2 500)	14 000 (2 600 - 32 000)
Faecal coliforms	counts/ 100mL	6 100 (920 - 52 000)	1 900 (330 - 7 300)	56 000 (8 100 - 140 000)
Ammonia-nitrogen	mg/L	0.05 (0.02 - 0.10)	0.01 (<0.01 - 0.05)	0.56 (0.25 - 1.60)
Nitrate-nitrogen	mg/L	0.77 (0.59 - 0.99)	0.46 (0.17 - 0.75)	1.10 (0.84 - 1.50)
Total Kjeldahl Nitrogen	mg/L	0.16 (0.10 - 0.37)	0.10 (0.06 - 0.34)	0.71 (0.51 - 2.00)
Ortho-phosphate Phosphorus	mg/L	0.03 (<0.01 - 0.06)	0.01 (<0.01 - 0.02)	0.09 (0.04 - 0.20)
Total phosphorus	mg/L	0.04 (0.03 - 0.08)	0.02 (<0.02 - 0.07)	0.16 (0.10 - 0.28)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	50 (<50 - 214)	<50 (<50 - 84)	69 (<50 - 217)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 3)	1 (<1 - 2)	1 (<1 - 9)
Lead	µg/L	<1 (<1 - 10)	<1 (<1 - 1)	<1 (<1 - 2)
Zinc	µg/L	15 (10 - 22)	22 (14 - 39)	14 (11 - 32)
Flow	L/s	50 (17 - 110)	30 (10 - 66)	92 (57 - 426)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River in 2016 (Part 2 of 3)

Parameter	Unit	Lam Tsuen River		
		TR12G	TR12F	TR12B
Dissolved oxygen	mg/L	8.3 (7.5 - 9.8)	8.6 (7.8 - 10.3)	8.8 (7.9 - 10.5)
pH		7.5 (7.1 - 7.7)	7.6 (7.4 - 7.8)	7.3 (7.1 - 7.5)
Suspended solids	mg/L	2 (<1 - 8)	2 (<1 - 52)	1 (<1 - 4)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - <1)	<1 (<1 - 1)	<1 (<1 - 1)
Chemical Oxygen Demand	mg/L	4 (<2 - 9)	4 (<2 - 10)	3 (<2 - 5)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	280 (100 - 1 500)	1 000 (340 - 6 800)	790 (150 - 5 100)
Faecal coliforms	counts/ 100mL	2 900 (340 - 25 000)	8 800 (1 100 - 160 000)	5 600 (800 - 47 000)
Ammonia-nitrogen	mg/L	0.02 (<0.01 - 0.03)	0.03 (0.02 - 0.10)	0.04 (0.03 - 0.07)
Nitrate-nitrogen	mg/L	0.16 (0.05 - 0.25)	0.49 (0.27 - 0.72)	0.82 (0.56 - 1.40)
Total Kjeldahl nitrogen	mg/L	0.14 (0.07 - 0.23)	0.25 (0.09 - 0.50)	0.16 (<0.05 - 0.33)
Ortho-phosphate Phosphorus	mg/L	0.02 (<0.01 - 0.04)	0.03 (<0.01 - 0.13)	0.04 (<0.01 - 0.07)
Total phosphorus	mg/L	0.05 (0.03 - 0.07)	0.08 (0.04 - 0.26)	0.06 (0.04 - 0.08)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 81)	<50 (<50 - 266)	54 (<50 - 80)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	1 (<1 - 3)	1 (<1 - 4)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - 9)	<1 (<1 - 1)
Zinc	µg/L	11 (<10 - 27)	13 (<10 - 45)	14 (<10 - 27)
Flow	L/s	28 (19 - 66)	54 (14 - 392)	181 (132 - 549)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Lam Tsuen River (Part 3 of 3) and Tai Po River in 2016

Parameter	Unit	Lam Tsuen River			Tai Po River
		TR12E	TR12	TR12I	TR13
Dissolved oxygen	mg/L	8.5 (7.9 - 10.0)	8.5 (7.7 - 10.0)	6.7 (5.0 - 8.5)	7.2 (5.5 - 8.2)
pH		7.6 (7.4 - 8.0)	7.9 (7.5 - 8.3)	7.4 (7.1 - 7.9)	7.4 (7.1 - 7.5)
Suspended solids	mg/L	9 (1 - 37)	6 (2 - 590)	2 (1 - 7)	2 (<1 - 14)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 2)	4 (2 - 13)	2 (<1 - 4)	<1 (<1 - 5)
Chemical Oxygen Demand	mg/L	4 (<2 - 13)	8 (4 - 16)	10 (4 - 19)	10 (4 - 12)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)
E. coli	counts/ 100mL	3 100 (350 - 10 000)	3 400 (600 - 23 000)	49 000 (6 200 - 330 000)	6 900 (1 500 - 16 000)
Faecal coliforms	counts/ 100mL	8 600 (1 100 - 110 000)	15 000 (4 600 - 450 000)	160 000 (27 000 - 840 000)	17 000 (2 600 - 54 000)
Ammonia-nitrogen	mg/L	0.05 (0.04 - 0.11)	1.05 (0.29 - 5.50)	0.64 (0.15 - 1.20)	0.19 (0.12 - 0.31)
Nitrate-nitrogen	mg/L	0.74 (0.42 - 1.10)	1.00 (0.56 - 1.60)	0.74 (0.25 - 1.10)	0.43 (0.25 - 0.86)
Total Kjeldahl nitrogen	mg/L	0.23 (0.19 - 0.59)	1.50 (0.53 - 7.20)	0.87 (0.36 - 1.30)	0.33 (0.26 - 0.52)
Ortho-phosphate Phosphorus	mg/L	0.02 (<0.01 - 0.03)	0.16 (0.03 - 0.38)	0.08 (0.01 - 0.13)	0.04 (0.02 - 0.06)
Total phosphorus	mg/L	0.05 (<0.02 - 0.20)	0.20 (0.09 - 0.53)	0.12 (0.06 - 0.23)	0.08 (0.04 - 0.09)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	430 (146 - 2 648)	123 (<50 - 1 181)	77 (<50 - 257)	54 (<50 - 69)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 3)	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	2 (<1 - 4)	2 (1 - 10)	2 (2 - 3)	2 (1 - 9)
Lead	µg/L	<1 (<1 - 3)	1 (<1 - 64)	<1 (<1 - 2)	<1 (<1 - <1)
Zinc	µg/L	20 (12 - 28)	21 (13 - 54)	19 (14 - 44)	17 (<10 - 27)
Flow	L/s	157 (96 - 212)	95 (16 - 232)	NM	114 (39 - 190)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Po Kau Stream, Shan Liu Stream and Tung Tze Stream in 2016

Parameter	Unit	Tai Po Kau Stream TR14	Shan Liu Stream TR4	Tung Tze Stream TR6
Dissolved oxygen	mg/L	8.3 (7.0 - 9.9)	8.1 (7.5 - 9.6)	5.9 (5.1 - 8.2)
pH		7.5 (7.1 - 7.9)	8.0 (7.4 - 8.8)	7.3 (7.1 - 7.8)
Suspended solids	mg/L	3 (<1 - 70)	5 (1 - 87)	5 (1 - 53)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - <1)	<1 (<1 - 2)	2 (<1 - 6)
Chemical Oxygen Demand	mg/L	4 (2 - 8)	4 (2 - 7)	14 (8 - 35)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 1.0)	<0.5 (<0.5 - 1.1)
E. coli	counts/ 100mL	250 (28 - 800)	3 200 (440 - 42 000)	1 800 (44 - 19 000)
Faecal coliforms	counts/ 100mL	980 (51 - 8 400)	10 000 (510 - 60 000)	4 400 (190 - 37 000)
Ammonia-nitrogen	mg/L	0.05 (0.02 - 0.12)	0.10 (0.03 - 0.23)	0.60 (0.08 - 1.60)
Nitrate-nitrogen	mg/L	0.23 (0.19 - 0.60)	0.57 (0.36 - 0.81)	0.18 (0.02 - 0.33)
Total Kjeldahl nitrogen	mg/L	0.22 (0.17 - 0.36)	0.31 (0.15 - 0.51)	0.93 (0.12 - 2.00)
Ortho-phosphate Phosphorus	mg/L	0.01 (<0.01 - 0.03)	0.06 (0.05 - 0.07)	0.06 (<0.01 - 0.15)
Total phosphorus	mg/L	0.03 (<0.02 - 0.04)	0.08 (0.06 - 0.12)	0.11 (0.06 - 0.25)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 61)	55 (<50 - 84)	<50 (<50 - 80)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 5)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	<1 (<1 - 3)	<1 (<1 - 2)	3 (2 - 4)
Lead	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	<1 (<1 - 1)
Zinc	µg/L	12 (<10 - 17)	12 (10 - 16)	13 (<10 - 20)
Flow	L/s	34 (12 - 123)	64 (48 - 105)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ho Chung River in 2016

Parameter	Unit	Ho Chung River	
		PR1	PR2
Dissolved oxygen	mg/L	7.1 (5.7 - 9.8)	8.3 (8.1 - 10.1)
pH		7.1 (6.9 - 7.3)	7.4 (7.2 - 7.8)
Suspended solids	mg/L	4 (<1 - 6)	3 (1 - 7)
5-day Biochemical Oxygen Demand	mg/L	2 (1 - 6)	<1 (<1 - 1)
Chemical Oxygen Demand	mg/L	11 (4 - 30)	3 (<2 - 7)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	30 000 (6 200 - 190 000)	2 400 (280 - 19 000)
Faecal coliforms	counts/ 100mL	72 000 (12 000 - 320 000)	12 000 (3 000 - 70 000)
Ammonia-nitrogen	mg/L	1.35 (0.30 - 3.70)	0.04 (<0.01 - 0.18)
Nitrate-nitrogen	mg/L	0.43 (0.12 - 0.67)	0.42 (0.22 - 0.51)
Total Kjeldahl nitrogen	mg/L	2.35 (0.89 - 5.00)	0.18 (0.12 - 0.46)
Ortho-phosphate Phosphorus	mg/L	0.12 (0.03 - 0.27)	0.02 (<0.01 - 0.03)
Total phosphorus	mg/L	0.28 (0.12 - 0.51)	0.03 (0.02 - 0.06)
Total sulphide	mg/L	<0.02 (<0.02 - 0.03)	<0.02 (<0.02 - <0.02)
Aluminium	μg/L	87 (56 - 138)	62 (<50 - 98)
Cadmium	μg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	μg/L	<1 (<1 - 3)	<1 (<1 - <1)
Copper	μg/L	2 (<1 - 4)	1 (<1 - 2)
Lead	μg/L	<1 (<1 - 1)	<1 (<1 - 1)
Zinc	μg/L	16 (12 - 30)	15 (<10 - 28)
Flow	L/s	NM	340 (100 - 810)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sha Kok Mei Stream in 2016

Parameter	Unit	Sha Kok Mei Stream PR5	Sha Kok Mei Stream PR6
Dissolved oxygen	mg/L	7.9 (7.3 - 9.4)	8.3 (8.0 - 10.2)
pH		7.6 (7.2 - 8.2)	7.4 (7.1 - 7.8)
Suspended solids	mg/L	2 (1 - 7)	3 (<1 - 38)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 4)	<1 (<1 - 5)
Chemical Oxygen Demand	mg/L	4 (3 - 7)	5 (2 - 11)
Oil & grease	mg/L	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	630 (<10 - 14 000)	14 000 (4 500 - 130 000)
Faecal coliforms	counts/ 100mL	2 600 (<10 - 43 000)	31 000 (8 100 - 140 000)
Ammonia-nitrogen	mg/L	0.17 (0.08 - 0.32)	0.15 (0.04 - 0.31)
Nitrate-nitrogen	mg/L	0.98 (0.63 - 1.30)	2.15 (1.70 - 2.80)
Total Kjeldahl nitrogen	mg/L	0.39 (0.18 - 0.92)	0.28 (0.05 - 0.87)
Ortho-phosphate Phosphorus	mg/L	0.05 (0.04 - 0.08)	0.07 (0.04 - 0.14)
Total phosphorus	mg/L	0.08 (0.06 - 0.12)	0.11 (0.05 - 0.23)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	55 (<50 - 141)	56 (<50 - 190)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 5)	2 (<1 - 36)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 5)
Zinc	µg/L	13 (<10 - 41)	14 (<10 - 39)
Flow	L/s	149 (80 - 404)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tai Chung Hau Stream in 2016

Parameter	Unit	Tai Chung Hau Stream	
		PR7	PR8
Dissolved oxygen	mg/L	8.2 (8.0 - 9.6)	8.0 (7.6 - 9.9)
pH		7.5 (7.3 - 8.7)	7.9 (7.3 - 9.5)
Suspended solids	mg/L	3 (1 - 72)	2 (1 - 18)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 4)	2 (<1 - 7)
Chemical Oxygen Demand	mg/L	5 (3 - 11)	6 (2 - 8)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	11 000 (4 400 - 97 000)	14 000 (5 100 - 84 000)
Faecal coliforms	counts/ 100mL	29 000 (8 700 - 130 000)	33 000 (10 000 - 170 000)
Ammonia-nitrogen	mg/L	0.09 (0.03 - 0.29)	0.12 (0.04 - 0.41)
Nitrate-nitrogen	mg/L	0.74 (0.51 - 1.20)	1.05 (0.70 - 1.70)
Total Kjeldahl Nitrogen	mg/L	0.35 (0.15 - 0.72)	0.39 (0.18 - 0.94)
Ortho-phosphate Phosphorus	mg/L	0.04 (0.03 - 0.06)	0.05 (0.03 - 0.13)
Total phosphorus	mg/L	0.08 (0.04 - 0.17)	0.10 (0.05 - 0.17)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	106 (<50 - 270)	87 (<50 - 257)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 4)
Copper	µg/L	2 (<1 - 5)	1 (<1 - 8)
Lead	µg/L	<1 (<1 - 8)	<1 (<1 - 2)
Zinc	µg/L	15 (10 - 60)	18 (<10 - 52)
Flow	L/s	133 (24 - 300)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tseng Lan Shue Stream in 2016

Parameter	Unit	Tseng Lan Shue Stream		
		JR3	JR6	JR11
Dissolved oxygen	mg/L	7.5 (5.2 - 9.1)	7.9 (7.4 - 9.3)	8.4 (8.0 - 10.2)
pH		7.3 (6.9 - 7.7)	7.5 (7.2 - 7.8)	7.7 (7.5 - 8.1)
Suspended solids	mg/L	5 (2 - 48)	3 (1 - 48)	2 (<1 - 5)
5-day Biochemical Oxygen Demand	mg/L	6 (2 - 13)	4 (2 - 20)	1 (<1 - 3)
Chemical Oxygen Demand	mg/L	13 (4 - 20)	11 (5 - 31)	6 (3 - 9)
Oil & grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.9)
E. coli	counts/ 100mL	39 000 (100 - 160 000)	83 000 (17 000 - 370 000)	2 000 (330 - 75 000)
Faecal coliforms	counts/ 100mL	71 000 (300 - 250 000)	170 000 (51 000 - 640 000)	7 700 (960 - 150 000)
Ammonia-nitrogen	mg/L	3.15 (0.50 - 8.00)	0.36 (0.02 - 0.77)	0.07 (0.02 - 0.25)
Nitrate-nitrogen	mg/L	1.70 (1.20 - 2.30)	2.55 (1.00 - 3.30)	2.45 (0.98 - 4.40)
Total Kjeldahl nitrogen	mg/L	4.20 (0.91 - 9.60)	1.05 (0.24 - 2.40)	0.32 (<0.05 - 0.61)
Ortho-phosphate Phosphorus	mg/L	0.39 (0.09 - 0.65)	0.35 (0.10 - 0.54)	0.27 (0.07 - 0.41)
Total phosphorus	mg/L	0.58 (0.27 - 0.90)	0.50 (0.14 - 0.73)	0.31 (0.09 - 0.45)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	161 (57 - 419)	102 (<50 - 450)	66 (<50 - 141)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - <1)
Copper	µg/L	2 (2 - 8)	3 (2 - 14)	2 (1 - 3)
Lead	µg/L	1 (<1 - 16)	1 (<1 - 10)	<1 (<1 - 2)
Zinc	µg/L	24 (15 - 65)	26 (21 - 108)	20 (12 - 51)
Flow	L/s	NM	NM	92 (44 - 368)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Indus in 2016

Parameter	Unit	River Indus IN1	IN2	IN3
Dissolved oxygen	mg/L	4.5 (3.2 - 6.8)	6.2 (5.4 - 9.7)	8.2 (7.2 - 9.7)
pH		7.2 (7.0 - 7.7)	7.3 (7.1 - 7.6)	7.6 (7.3 - 9.9)
Suspended solids	mg/L	12 (6 - 63)	6 (2 - 16)	7 (2 - 100)
5-day Biochemical Oxygen Demand	mg/L	4 (3 - 20)	2 (<1 - 7)	<1 (<1 - 3)
Chemical Oxygen Demand	mg/L	16 (8 - 46)	9 (5 - 14)	6 (3 - 13)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	130 000 (2 500 - 1 800 000)	9 000 (1 600 - 130 000)	2 800 (900 - 6 100)
Faecal coliforms	counts/ 100mL	330 000 (17 000 - 3 800 000)	44 000 (9 200 - 1 300 000)	11 000 (4 300 - 44 000)
Ammonia-nitrogen	mg/L	1.80 (0.08 - 8.80)	0.64 (0.24 - 1.40)	0.09 (0.03 - 0.79)
Nitrate-nitrogen	mg/L	2.35 (0.65 - 3.50)	1.30 (0.68 - 1.50)	0.75 (0.53 - 1.30)
Total Kjeldahl nitrogen	mg/L	3.05 (0.40 - 12.00)	0.96 (0.59 - 2.90)	0.42 (0.23 - 1.70)
Ortho-phosphate Phosphorus	mg/L	0.24 (0.16 - 0.41)	0.09 (0.04 - 0.16)	0.09 (0.04 - 0.16)
Total phosphorus	mg/L	0.40 (0.27 - 1.10)	0.16 (0.10 - 0.30)	0.13 (0.08 - 0.29)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	μg/L	124 (74 - 380)	109 (65 - 252)	204 (116 - 802)
Cadmium	μg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	μg/L	<1 (<1 - 4)	<1 (<1 - <1)	2 (<1 - 7)
Copper	μg/L	3 (2 - 9)	3 (2 - 10)	2 (1 - 6)
Lead	μg/L	1 (<1 - 5)	1 (<1 - 4)	1 (<1 - 4)
Zinc	μg/L	22 (14 - 42)	29 (19 - 41)	17 (<10 - 38)
Flow	L/s	NM	NM	19 (15 - 42)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Beas in 2016

Parameter	Unit	River Beas RB1	River Beas RB2	River Beas RB3
Dissolved oxygen	mg/L	9.0 (8.0 - 10.4)	7.6 (6.7 - 9.9)	6.9 (3.4 - 9.9)
pH		7.6 (7.4 - 7.7)	7.5 (7.2 - 7.6)	7.4 (7.2 - 7.6)
Suspended solids	mg/L	4 (1 - 10)	4 (2 - 8)	6 (3 - 23)
5-day Biochemical Oxygen Demand	mg/L	2 (<1 - 5)	3 (1 - 11)	4 (2 - 11)
Chemical Oxygen Demand	mg/L	6 (4 - 20)	8 (5 - 15)	12 (7 - 28)
Oil & grease	mg/L	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	6 500 (1 700 - 34 000)	3 300 (760 - 8 000)	21 000 (4 500 - 150 000)
Faecal coliforms	counts/ 100mL	36 000 (4 000 - 440 000)	17 000 (1 200 - 58 000)	78 000 (22 000 - 440 000)
Ammonia-nitrogen	mg/L	0.19 (0.08 - 5.60)	0.67 (0.32 - 1.60)	1.45 (0.02 - 3.10)
Nitrate-nitrogen	mg/L	0.89 (0.57 - 1.30)	0.70 (0.14 - 1.10)	0.98 (0.63 - 1.70)
Total Kjeldahl nitrogen	mg/L	0.86 (0.31 - 6.70)	1.35 (0.43 - 2.70)	2.05 (0.25 - 4.90)
Ortho-phosphate Phosphorus	mg/L	0.17 (0.08 - 0.33)	0.11 (0.04 - 0.24)	0.19 (<0.01 - 0.38)
Total phosphorus	mg/L	0.27 (0.12 - 0.63)	0.28 (0.12 - 0.47)	0.35 (0.13 - 0.75)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	59 (<50 - 95)	52 (<50 - 123)	95 (<50 - 220)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 2)
Copper	µg/L	2 (1 - 4)	2 (1 - 4)	3 (2 - 7)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 1)	1 (<1 - 3)
Zinc	µg/L	17 (13 - 50)	21 (15 - 29)	57 (25 - 78)
Flow	L/s	188 (52 - 284)	228 (80 - 1 020)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for River Ganges in 2016

Parameter	Unit	River Ganges GR1	River Ganges GR2	River Ganges GR3
Dissolved oxygen	mg/L	7.9 (6.3 - 10.0)	6.9 (6.2 - 9.2)	8.0 (7.4 - 9.6)
pH		7.3 (7.2 - 7.6)	7.3 (7.0 - 7.5)	7.4 (7.1 - 7.8)
Suspended solids	mg/L	12 (3 - 23)	13 (2 - 33)	3 (1 - 34)
5-day Biochemical Oxygen Demand	mg/L	4 (2 - 15)	1 (<1 - 9)	<1 (<1 - 5)
Chemical Oxygen Demand	mg/L	11 (7 - 19)	9 (6 - 25)	4 (<2 - 17)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	12 000 (2 900 - 91 000)	13 000 (2 200 - 66 000)	600 (25 - 12 000)
Faecal coliforms	counts/ 100mL	32 000 (10 000 - 180 000)	41 000 (10 000 - 110 000)	5 400 (200 - 880 000)
Ammonia-nitrogen	mg/L	1.80 (0.17 - 5.60)	0.24 (0.15 - 0.72)	0.05 (0.02 - 0.18)
Nitrate-nitrogen	mg/L	1.30 (1.00 - 2.30)	0.80 (0.60 - 2.30)	0.26 (0.15 - 0.78)
Total Kjeldahl nitrogen	mg/L	2.20 (0.54 - 8.30)	0.79 (0.45 - 2.40)	0.45 (0.22 - 0.72)
Ortho-phosphate Phosphorus	mg/L	0.56 (0.20 - 1.30)	0.18 (0.05 - 0.43)	<0.01 (<0.01 - 0.04)
Total phosphorus	mg/L	0.76 (0.33 - 1.70)	0.35 (0.18 - 0.69)	0.05 (<0.02 - 0.25)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	110 (52 - 198)	99 (59 - 254)	60 (<50 - 430)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	3 (1 - 11)	3 (2 - 12)	1 (<1 - 5)
Lead	µg/L	1 (<1 - 3)	2 (<1 - 8)	<1 (<1 - 5)
Zinc	µg/L	24 (18 - 45)	28 (19 - 55)	17 (10 - 56)
Flow	L/s	26 (12 - 37)	29 (20 - 47)	90 (45 - 105)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2016 (Part 1 of 2)

Parameter	Unit	Yuen Long Creek	
		YL1	YL2
Dissolved oxygen	mg/L	6.8 (4.1 - 8.4)	5.5 (3.5 - 7.5)
pH		7.4 (7.2 - 7.7)	7.3 (7.1 - 7.5)
Suspended solids	mg/L	9 (4 - 35)	8 (2 - 46)
5-day Biochemical Oxygen Demand	mg/L	8 (4 - 65)	11 (3 - 42)
Chemical Oxygen Demand	mg/L	17 (9 - 56)	37 (13 - 130)
Oil & grease	mg/L	<0.5 (<0.5 - 0.8)	0.6 (<0.5 - 2.8)
E. coli	counts/ 100mL	140 000 (32 000 - 310 000)	100 000 (31 000 - 600 000)
Faecal coliforms	counts/ 100mL	310 000 (75 000 - 900 000)	240 000 (56 000 - 2 400 000)
Ammonia-nitrogen	mg/L	2.80 (0.70 - 9.80)	10.35 (0.77 - 21.00)
Nitrate-nitrogen	mg/L	0.76 (<0.01 - 1.10)	0.89 (<0.01 - 4.80)
Total Kjeldahl nitrogen	mg/L	6.15 (1.60 - 18.00)	13.50 (1.50 - 25.00)
Ortho-phosphate Phosphorus	mg/L	0.44 (0.27 - 1.10)	2.15 (0.20 - 3.00)
Total phosphorus	mg/L	0.73 (0.45 - 1.60)	2.50 (0.33 - 3.80)
Total sulphide	mg/L	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 0.06)
Aluminium	µg/L	201 (98 - 654)	102 (55 - 273)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 1)
Copper	µg/L	5 (3 - 20)	4 (3 - 15)
Lead	µg/L	2 (<1 - 15)	1 (<1 - 5)
Zinc	µg/L	36 (22 - 97)	35 (19 - 78)
Flow	L/s	264 (78 - 949)	22 (9 - 66)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Yuen Long Creek in 2016 (Part 2 of 2)

Parameter	Unit	Yuen Long Creek	
		YL3	YL4
Dissolved oxygen	mg/L	4.7 (2.8 - 8.3)	3.8 (2.4 - 8.2)
pH		7.6 (7.3 - 8.1)	7.4 (7.1 - 8.2)
Suspended solids	mg/L	19 (8 - 220)	33 (13 - 58)
5-day Biochemical Oxygen Demand	mg/L	27 (12 - 77)	84 (17 - 170)
Chemical Oxygen Demand	mg/L	28 (10 - 95)	74 (12 - 140)
Oil & grease	mg/L	0.6 (<0.5 - 1.4)	0.8 (<0.5 - 11.0)
E. coli	counts/ 100mL	460 000 (240 000 - 770 000)	1 300 000 (320 000 - 3 700 000)
Faecal coliforms	counts/ 100mL	1 200 000 (510 000 - 3 200 000)	3 300 000 (680 000 - 9 100 000)
Ammonia-nitrogen	mg/L	3.55 (1.20 - 12.00)	5.05 (0.18 - 7.00)
Nitrate-nitrogen	mg/L	0.30 (<0.01 - 1.30)	<0.01 (<0.01 - 0.79)
Total Kjeldahl nitrogen	mg/L	4.95 (2.00 - 17.00)	9.90 (2.20 - 12.00)
Ortho-phosphate Phosphorus	mg/L	0.32 (0.17 - 0.78)	0.31 (0.07 - 0.68)
Total phosphorus	mg/L	0.67 (0.28 - 1.90)	1.20 (0.37 - 1.40)
Total sulphide	mg/L	0.03 (<0.02 - 0.05)	0.16 (<0.02 - 0.48)
Aluminium	µg/L	156 (121 - 475)	223 (87 - 463)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 3)
Copper	µg/L	6 (2 - 11)	7 (4 - 11)
Lead	µg/L	2 (1 - 11)	2 (1 - 8)
Zinc	µg/L	40 (21 - 110)	48 (29 - 92)
Flow	L/s	645 (270 - 2 003)	105 (84 - 456)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kam Tin River in 2016

Parameter	Unit	Kam Tin River	
		KT1	KT2
Dissolved oxygen	mg/L	5.9 (4.4 - 8.0)	5.0 (3.2 - 8.6)
pH		7.4 (7.0 - 7.8)	7.4 (7.0 - 7.6)
Suspended solids	mg/L	16 (4 - 33)	12 (3 - 33)
5-day Biochemical Oxygen Demand	mg/L	12 (4 - 51)	13 (4 - 35)
Chemical Oxygen Demand	mg/L	17 (10 - 57)	18 (11 - 44)
Oil & grease	mg/L	0.5 (<0.5 - 2.1)	<0.5 (<0.5 - 1.3)
E. coli	counts/ 100mL	320 000 (45 000 - 3 100 000)	140 000 (28 000 - 660 000)
Faecal coliforms	counts/ 100mL	630 000 (150 000 - 3 500 000)	270 000 (55 000 - 850 000)
Ammonia-nitrogen	mg/L	5.20 (2.00 - 8.00)	4.65 (0.85 - 13.00)
Nitrate-nitrogen	mg/L	0.75 (<0.01 - 1.10)	0.37 (<0.01 - 0.83)
Total Kjeldahl nitrogen	mg/L	6.05 (2.50 - 14.00)	5.45 (1.60 - 18.00)
Ortho-phosphate Phosphorus	mg/L	0.89 (0.45 - 3.00)	0.73 (0.34 - 2.30)
Total phosphorus	mg/L	1.30 (0.60 - 3.80)	0.98 (0.44 - 3.20)
Total sulphide	mg/L	0.03 (<0.02 - 0.04)	<0.02 (<0.02 - 0.05)
Aluminium	µg/L	85 (54 - 328)	99 (60 - 375)
Cadmium	µg/L	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - 1)
Copper	µg/L	15 (6 - 29)	5 (2 - 10)
Lead	µg/L	1 (<1 - 8)	<1 (<1 - 4)
Zinc	µg/L	51 (34 - 104)	30 (16 - 60)
Flow	L/s	409 (137 - 998)	386 (90 - 984)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tin Shui Wai Nullah and Fairview Park Nullah in 2016

Parameter	Unit	Tin Shui Wai Nullah		Fairview Park Nullah
		TSR1	TSR2	FVR1
Dissolved oxygen	mg/L	7.2 (3.7 - 10.4)	9.1 (6.7 - 11.2)	5.5 (4.2 - 8.9)
pH		7.5 (7.2 - 8.0)	7.7 (7.4 - 8.6)	7.3 (7.1 - 7.5)
Suspended solids	mg/L	6 (2 - 160)	4 (1 - 50)	22 (6 - 46)
5-day Biochemical Oxygen Demand	mg/L	8 (1 - 70)	1 (<1 - 8)	5 (2 - 12)
Chemical Oxygen Demand	mg/L	12 (5 - 28)	5 (2 - 14)	18 (8 - 33)
Oil & grease	mg/L	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 0.8)	<0.5 (<0.5 - 1.0)
E. coli	counts/ 100mL	47 000 (1 700 - 390 000)	21 000 (5 800 - 48 000)	44 000 (3 800 - 5 100 000)
Faecal coliforms	counts/ 100mL	210 000 (10 000 - 2 000 000)	56 000 (19 000 - 310 000)	120 000 (11 000 - 7 500 000)
Ammonia-nitrogen	mg/L	1.30 (0.78 - 4.30)	0.33 (0.06 - 1.00)	2.60 (0.95 - 5.10)
Nitrate-nitrogen	mg/L	0.78 (0.17 - 1.20)	0.66 (0.53 - 0.85)	0.67 (0.36 - 1.20)
Total Kjeldahl nitrogen	mg/L	1.90 (1.20 - 6.80)	0.68 (0.17 - 1.50)	3.85 (1.60 - 5.30)
Ortho-phosphate Phosphorus	mg/L	0.13 (0.08 - 0.42)	0.04 (<0.01 - 0.13)	0.33 (0.20 - 0.64)
Total phosphorus	mg/L	0.25 (0.16 - 1.30)	0.08 (0.04 - 0.15)	0.63 (0.38 - 0.90)
Total sulphide	mg/L	<0.02 (<0.02 - 0.13)	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	146 (59 - 2 270)	140 (77 - 589)	171 (66 - 299)
Cadmium	µg/L	<0.1 (<0.1 - 0.6)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 8)	<1 (<1 - 2)	<1 (<1 - 2)
Copper	µg/L	3 (2 - 15)	1 (<1 - 15)	3 (2 - 18)
Lead	µg/L	1 (<1 - 36)	<1 (<1 - 14)	2 (<1 - 3)
Zinc	µg/L	29 (15 - 361)	17 (11 - 118)	31 (15 - 232)
Flow	L/s	NM	50 (33 - 195)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ha Pak Nai Stream, Pak Nai Stream and Sheung Pak Nai Stream in 2016

Parameter	Unit	Ha Pak Nai Stream DB1	Pak Nai Stream DB3	Sheung Pak Nai Stream DB5
Dissolved oxygen	mg/L	8.5 (7.7 - 10.1)	8.2 (7.5 - 9.9)	8.1 (7.6 - 9.7)
pH		7.6 (7.4 - 8.3)	7.3 (7.0 - 7.9)	7.2 (7.0 - 7.9)
Suspended solids	mg/L	2 (<1 - 88)	5 (1 - 210)	16 (5 - 420)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	<1 (<1 - 1)	1 (<1 - 4)
Chemical Oxygen Demand	mg/L	<2 (<2 - 9)	<2 (<2 - 12)	5 (3 - 9)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - 0.7)
E. coli	counts/ 100mL	94 (13 - 3 700)	440 (150 - 4 900)	6 700 (510 - 56 000)
Faecal coliforms	counts/ 100mL	970 (35 - 6 100)	2 100 (210 - 26 000)	13 000 (2 400 - 58 000)
Ammonia-nitrogen	mg/L	0.01 (<0.01 - 0.12)	0.02 (<0.01 - 0.09)	0.17 (0.04 - 1.10)
Nitrate-nitrogen	mg/L	0.32 (0.23 - 0.50)	0.26 (0.21 - 0.39)	0.29 (0.23 - 0.55)
Total Kjeldahl nitrogen	mg/L	0.11 (0.05 - 0.38)	0.11 (0.08 - 0.61)	0.65 (0.13 - 2.50)
Ortho-phosphate Phosphorus	mg/L	<0.01 (<0.01 - <0.01)	<0.01 (<0.01 - 0.02)	<0.01 (<0.01 - 0.16)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.05)	<0.02 (<0.02 - 0.06)	0.06 (<0.02 - 0.25)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	59 (<50 - 375)	50 (<50 - 336)	68 (<50 - 1 298)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - 1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 5)	2 (<1 - 24)
Lead	µg/L	<1 (<1 - 7)	1 (<1 - 15)	2 (1 - 93)
Zinc	µg/L	10 (<10 - 16)	13 (<10 - 25)	13 (<10 - 43)
Flow	L/s	27 (11 - 53)	39 (15 - 59)	29 (21 - 57)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Ngau Hom Sha Stream, Tai Shui Hang Stream and Tsang Kok Stream in 2016

Parameter	Unit	Ngau Hom Sha Stream DB6	Tai Shui Hang Stream DB2	Tsang Kok Stream DB8
Dissolved oxygen	mg/L	8.0 (7.1 - 9.6)	8.5 (7.7 - 10.0)	8.8 (7.9 - 10.1)
pH		7.0 (6.7 - 7.4)	7.8 (7.4 - 8.4)	7.7 (7.4 - 8.7)
Suspended solids	mg/L	7 (2 - 160)	2 (<1 - 260)	4 (<1 - 230)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 2)	<1 (<1 - 2)	<1 (<1 - 4)
Chemical Oxygen Demand	mg/L	4 (<2 - 11)	<2 (<2 - 15)	4 (<2 - 20)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.5)
E. coli	counts/ 100mL	1 300 (230 - 50 000)	160 (27 - 6 800)	650 (110 - 8 400)
Faecal coliforms	counts/ 100mL	7 700 (440 - 630 000)	1 500 (59 - 20 000)	5 700 (660 - 48 000)
Ammonia-nitrogen	mg/L	0.15 (0.04 - 0.26)	0.01 (<0.01 - 0.11)	0.13 (0.02 - 8.90)
Nitrate-nitrogen	mg/L	0.44 (0.32 - 0.67)	0.31 (0.20 - 0.49)	1.65 (1.00 - 2.30)
Total Kjeldahl nitrogen	mg/L	0.29 (0.19 - 0.69)	0.12 (0.07 - 0.70)	0.28 (0.09 - 11.00)
Ortho-phosphate Phosphorus	mg/L	0.03 (<0.01 - 0.05)	<0.01 (<0.01 - 0.01)	<0.01 (<0.01 - 0.13)
Total phosphorus	mg/L	0.08 (0.05 - 0.11)	<0.02 (<0.02 - 0.06)	0.03 (<0.02 - 0.20)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	60 (<50 - 622)	67 (<50 - 596)	77 (<50 - 2 237)
Cadmium	µg/L	<0.1 (<0.1 - 0.8)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.7)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	2 (1 - 17)	<1 (<1 - 5)	1 (<1 - 7)
Lead	µg/L	2 (<1 - 21)	1 (<1 - 22)	1 (<1 - 89)
Zinc	µg/L	40 (18 - 270)	12 (<10 - 30)	16 (<10 - 94)
Flow	L/s	22 (6 - 42)	180 (100 - 520)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2016 (Part 1 of 2)

Parameter	Unit	Mui Wo River		
		MW1	MW2	MW3
Dissolved oxygen	mg/L	8.5 (7.5 - 9.9)	8.8 (7.5 - 9.7)	8.7 (8.3 - 10.7)
pH		7.4 (7.2 - 8.1)	7.3 (7.1 - 7.8)	7.6 (7.3 - 8.4)
Suspended solids	mg/L	2 (<1 - 3)	3 (1 - 8)	<1 (<1 - 2)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Chemical Oxygen Demand	mg/L	3 (<2 - 7)	7 (3 - 10)	<2 (<2 - 6)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	610 (280 - 1 200)	1 200 (460 - 4 000)	230 (58 - 1 100)
Faecal coliforms	counts/ 100mL	4 900 (1 100 - 17 000)	6 800 (4 100 - 10 000)	2 100 (750 - 5 700)
Ammonia-nitrogen	mg/L	0.05 (0.01 - 0.10)	0.08 (0.02 - 0.33)	0.01 (<0.01 - 0.03)
Nitrate-nitrogen	mg/L	0.41 (0.20 - 0.73)	0.24 (0.17 - 0.58)	0.40 (0.25 - 0.66)
Total Kjeldahl nitrogen	mg/L	0.21 (0.07 - 0.32)	0.32 (0.11 - 0.53)	0.14 (0.07 - 0.19)
Ortho-phosphate Phosphorus	mg/L	0.06 (<0.01 - 0.08)	0.03 (0.01 - 0.11)	0.03 (<0.01 - 0.06)
Total phosphorus	mg/L	0.08 (0.03 - 0.12)	0.06 (0.05 - 0.13)	0.05 (<0.02 - 0.07)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	61 (<50 - 112)	50 (<50 - 118)	52 (<50 - 90)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 3)	1 (<1 - 35)	<1 (<1 - 14)
Lead	µg/L	<1 (<1 - <1)	<1 (<1 - 2)	<1 (<1 - <1)
Zinc	µg/L	17 (10 - 29)	14 (<10 - 26)	13 (11 - 23)
Flow	L/s	36 (8 - 330)	NM	32 (5 - 159)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Mui Wo River in 2016 (Part 2 of 2)

Parameter	Unit	Mui Wo River	
		MW4	MW5
Dissolved oxygen	mg/L	8.2 (6.7 - 9.1)	8.0 (7.4 - 9.7)
pH		7.0 (6.7 - 7.4)	7.3 (6.9 - 7.6)
Suspended solids	mg/L	4 (1 - 13)	5 (<1 - 13)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	1 (<1 - 2)
Chemical Oxygen Demand	mg/L	7 (4 - 11)	7 (3 - 13)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.8)
E. coli	counts/ 100mL	700 (160 - 13 000)	7 200 (2 800 - 24 000)
Faecal coliforms	counts/ 100mL	4 100 (740 - 27 000)	26 000 (10 000 - 76 000)
Ammonia-nitrogen	mg/L	0.15 (0.04 - 0.32)	0.30 (0.12 - 0.97)
Nitrate-nitrogen	mg/L	0.34 (0.09 - 0.65)	0.27 (0.17 - 0.61)
Total Kjeldahl nitrogen	mg/L	0.26 (0.15 - 0.64)	0.61 (0.25 - 1.30)
Ortho-phosphate Phosphorus	mg/L	0.03 (<0.01 - 0.08)	0.04 (0.01 - 0.10)
Total phosphorus	mg/L	0.10 (0.04 - 0.17)	0.10 (0.06 - 0.18)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	64 (<50 - 173)	70 (50 - 160)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 6)	<1 (<1 - 2)
Lead	µg/L	<1 (<1 - 1)	1 (<1 - 3)
Zinc	µg/L	18 (11 - 27)	15 (13 - 23)
Flow	L/s	78 (15 - 195)	38 (8 - 120)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tung Chung River in 2016

Parameter	Unit	Tung Chung River		
		TC1	TC2	TC3
Dissolved oxygen	mg/L	8.1 (7.4 - 9.4)	8.3 (7.3 - 10.0)	8.4 (7.8 - 9.9)
pH		7.3 (6.7 - 7.9)	7.3 (7.1 - 7.6)	7.9 (7.2 - 8.3)
Suspended solids	mg/L	<1 (<1 - 7)	2 (<1 - 8)	2 (1 - 7)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - <1)	<1 (<1 - 1)	4 (1 - 8)
Chemical Oxygen Demand	mg/L	<2 (<2 - 6)	4 (3 - 6)	5 (2 - 6)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	79 (22 - 780)	630 (170 - 2 600)	8 500 (2 100 - 60 000)
Faecal coliforms	counts/ 100mL	1 100 (170 - 3 500)	7 000 (3 200 - 99 000)	44 000 (8 400 - 100 000)
Ammonia-nitrogen	mg/L	0.01 (<0.01 - 0.04)	0.01 (<0.01 - 0.04)	0.20 (0.05 - 0.94)
Nitrate-nitrogen	mg/L	0.15 (0.07 - 0.31)	0.05 (<0.01 - 0.23)	0.15 (0.08 - 0.37)
Total Kjeldahl nitrogen	mg/L	0.10 (0.06 - 0.30)	0.16 (0.07 - 0.38)	0.50 (0.14 - 1.20)
Ortho-phosphate Phosphorus	mg/L	<0.01 (<0.01 - 0.01)	<0.01 (<0.01 - 0.01)	0.02 (<0.01 - 0.07)
Total phosphorus	mg/L	<0.02 (<0.02 - 0.04)	<0.02 (<0.02 - 0.04)	0.05 (0.02 - 0.14)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	<50 (<50 - 100)	<50 (<50 - 107)	<50 (<50 - 82)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	<1 (<1 - 3)
Lead	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	<1 (<1 - <1)
Zinc	µg/L	12 (<10 - 25)	13 (<10 - 21)	12 (<10 - 27)
Flow	L/s	46 (16 - 260)	125 (44 - 214)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2016 (Part 1 of 2)

Parameter	Unit	Tuen Mun River		
		TN1	TN2	TN3
Dissolved oxygen	mg/L	6.0 (3.9 - 10.3)	8.4 (7.5 - 11.3)	5.2 (4.4 - 8.0)
pH		7.9 (7.4 - 9.3)	7.7 (7.4 - 8.2)	7.5 (7.3 - 8.2)
Suspended solids	mg/L	7 (2 - 22)	4 (1 - 160)	4 (1 - 12)
5-day Biochemical Oxygen Demand	mg/L	17 (4 - 23)	3 (2 - 9)	2 (<1 - 9)
Chemical Oxygen Demand	mg/L	20 (8 - 36)	6 (4 - 17)	12 (7 - 18)
Oil & grease	mg/L	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - <0.5)
E. coli	counts/ 100mL	62 000 (2 600 - 190 000)	35 000 (11 000 - 100 000)	8 500 (30 - 90 000)
Faecal coliforms	counts/ 100mL	350 000 (80 000 - 3 200 000)	81 000 (14 000 - 330 000)	62 000 (490 - 850 000)
Ammonia-nitrogen	mg/L	5.50 (0.48 - 7.50)	1.10 (0.43 - 4.10)	0.46 (0.25 - 0.95)
Nitrate-nitrogen	mg/L	0.86 (0.30 - 1.40)	1.75 (1.00 - 4.80)	0.48 (0.09 - 0.79)
Total Kjeldahl nitrogen	mg/L	6.40 (1.00 - 8.40)	1.40 (0.47 - 4.80)	0.72 (0.26 - 1.20)
Ortho-phosphate Phosphorus	mg/L	0.48 (0.19 - 0.63)	0.15 (0.07 - 0.47)	0.04 (0.02 - 0.08)
Total phosphorus	mg/L	0.67 (0.28 - 1.00)	0.22 (0.10 - 0.60)	0.11 (0.07 - 0.25)
Total sulphide	mg/L	0.03 (<0.02 - 0.05)	<0.02 (<0.02 - 0.07)	<0.02 (<0.02 - 0.18)
Aluminium	µg/L	128 (61 - 386)	162 (63 - 1 268)	85 (<50 - 158)
Cadmium	µg/L	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.5)
Chromium	µg/L	<1 (<1 - 3)	<1 (<1 - 4)	<1 (<1 - 3)
Copper	µg/L	3 (2 - 11)	2 (1 - 18)	3 (2 - 5)
Lead	µg/L	1 (<1 - 8)	2 (<1 - 37)	<1 (<1 - 4)
Zinc	µg/L	23 (16 - 116)	19 (15 - 160)	21 (<10 - 47)
Flow	L/s	131 (41 - 286)	23 (13 - 84)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Tuen Mun River in 2016 (Part 2 of 2)

Parameter	Unit	Tuen Mun River		
		TN4	TN5	TN6
Dissolved oxygen	mg/L	6.1 (5.2 - 8.4)	5.8 (5.0 - 8.3)	5.3 (3.7 - 7.1)
pH		7.5 (7.2 - 7.8)	7.5 (7.3 - 7.8)	7.2 (7.1 - 7.7)
Suspended solids	mg/L	5 (<1 - 25)	5 (1 - 10)	3 (<1 - 11)
5-day Biochemical Oxygen Demand	mg/L	1 (<1 - 6)	1 (<1 - 10)	2 (<1 - 4)
Chemical Oxygen Demand	mg/L	12 (7 - 19)	10 (6 - 21)	11 (7 - 21)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	13 000 (300 - 80 000)	6 200 (<100 - 54 000)	8 200 (<10 - 210 000)
Faecal coliforms	counts/ 100mL	65 000 (2 700 - 550 000)	34 000 (700 - 400 000)	55 000 (1 100 - 1 000 000)
Ammonia-nitrogen	mg/L	0.47 (0.32 - 1.00)	0.44 (0.14 - 0.85)	0.61 (0.29 - 0.78)
Nitrate-nitrogen	mg/L	0.54 (0.09 - 0.96)	0.63 (0.15 - 0.97)	0.41 (0.13 - 0.81)
Total Kjeldahl nitrogen	mg/L	0.74 (0.33 - 1.60)	0.61 (0.35 - 1.40)	0.79 (0.34 - 1.50)
Ortho-phosphate Phosphorus	mg/L	0.04 (0.02 - 0.07)	0.03 (<0.01 - 0.07)	0.04 (0.02 - 0.08)
Total phosphorus	mg/L	0.11 (0.07 - 0.14)	0.11 (0.07 - 0.14)	0.11 (0.07 - 0.18)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.03)
Aluminium	µg/L	100 (<50 - 221)	90 (<50 - 180)	61 (<50 - 132)
Cadmium	µg/L	<0.1 (<0.1 - 0.3)	<0.1 (<0.1 - 0.2)	<0.1 (<0.1 - 0.2)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - 2)	<1 (<1 - 2)
Copper	µg/L	3 (2 - 12)	3 (2 - 7)	3 (2 - 7)
Lead	µg/L	<1 (<1 - 3)	<1 (<1 - 12)	<1 (<1 - 3)
Zinc	µg/L	21 (10 - 40)	16 (<10 - 168)	22 (11 - 40)
Flow	L/s	NM	NM	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Pai Min Kok Stream and Kau Wa Keng Stream in 2016

Parameter	Unit	Pai Min Kok Stream		Kau Wa Keng Stream
		AN1	AN2	KW3
Dissolved oxygen	mg/L	8.5 (8.0 - 11.7)	8.6 (8.0 - 11.6)	8.5 (7.5 - 10.8)
pH		7.6 (7.5 - 7.8)	7.8 (7.5 - 8.4)	7.5 (7.1 - 7.8)
Suspended solids	mg/L	4 (1 - 12)	5 (1 - 8)	4 (<1 - 47)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 3)	<1 (<1 - 1)	4 (<1 - 13)
Chemical Oxygen Demand	mg/L	5 (<2 - 12)	5 (3 - 16)	8 (3 - 23)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	8 400 (2 300 - 43 000)	5 000 (600 - 260 000)	53 000 (10 000 - 250 000)
Faecal coliforms	counts/ 100mL	23 000 (2 800 - 120 000)	15 000 (2 500 - 260 000)	100 000 (19 000 - 620 000)
Ammonia-nitrogen	mg/L	0.06 (0.03 - 0.32)	0.03 (0.01 - 0.22)	0.74 (0.09 - 1.60)
Nitrate-nitrogen	mg/L	0.75 (0.58 - 2.50)	0.51 (0.36 - 1.20)	2.00 (1.10 - 3.00)
Total Kjeldahl nitrogen	mg/L	0.42 (0.15 - 2.70)	0.27 (0.08 - 0.80)	1.10 (0.28 - 2.10)
Ortho-phosphate Phosphorus	mg/L	0.05 (<0.01 - 0.09)	0.04 (0.01 - 0.05)	0.09 (0.03 - 0.14)
Total phosphorus	mg/L	0.08 (0.04 - 0.11)	0.07 (0.04 - 0.10)	0.15 (0.04 - 0.34)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	92 (73 - 191)	95 (<50 - 264)	136 (57 - 589)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	1.0 (0.2 - 2.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - <1)	<1 (<1 - 6)
Copper	µg/L	6 (2 - 26)	2 (1 - 6)	4 (2 - 30)
Lead	µg/L	1 (<1 - 3)	1 (<1 - 3)	2 (1 - 20)
Zinc	µg/L	28 (16 - 59)	21 (13 - 42)	113 (56 - 303)
Flow	L/s	NM	9 (3 - 28)	30 (10 - 74)

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Sam Dip Tam Stream in 2016

Parameter	Unit	Sam Dip Tam Stream		
		TW1	TW2	TW3
Dissolved oxygen	mg/L	8.2 (7.8 - 10.9)	8.6 (8.0 - 11.1)	8.5 (7.9 - 11.3)
pH		7.6 (7.4 - 7.9)	7.6 (7.4 - 7.9)	7.5 (7.4 - 8.4)
Suspended solids	mg/L	3 (1 - 6)	2 (<1 - 4)	1 (<1 - 6)
5-day Biochemical Oxygen Demand	mg/L	<1 (<1 - 1)	<1 (<1 - 3)	<1 (<1 - 2)
Chemical Oxygen Demand	mg/L	3 (<2 - 6)	4 (<2 - 7)	4 (<2 - 8)
Oil & grease	mg/L	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.7)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	1 700 (120 - 13 000)	7 600 (1 900 - 17 000)	4 600 (520 - 34 000)
Faecal coliforms	counts/ 100mL	9 600 (230 - 90 000)	31 000 (6 700 - 80 000)	15 000 (830 - 110 000)
Ammonia-nitrogen	mg/L	0.01 (<0.01 - 0.05)	0.08 (0.05 - 0.30)	0.06 (0.03 - 0.33)
Nitrate-nitrogen	mg/L	1.15 (0.89 - 1.70)	1.45 (1.00 - 1.70)	1.65 (1.20 - 2.10)
Total Kjeldahl nitrogen	mg/L	0.19 (<0.05 - 0.38)	0.33 (0.10 - 0.69)	0.24 (0.11 - 0.72)
Ortho-phosphate Phosphorus	mg/L	0.05 (0.03 - 0.06)	0.08 (0.06 - 0.13)	0.10 (0.09 - 0.17)
Total phosphorus	mg/L	0.06 (0.05 - 0.08)	0.11 (0.07 - 0.17)	0.12 (0.10 - 0.20)
Total sulphide	mg/L	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	62 (<50 - 137)	52 (<50 - 126)	<50 (<50 - 143)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - <1)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	1 (<1 - 4)	2 (1 - 8)	2 (1 - 10)
Lead	µg/L	1 (<1 - 2)	<1 (<1 - 3)	<1 (<1 - 2)
Zinc	µg/L	19 (<10 - 32)	18 (12 - 32)	17 (11 - 25)
Flow	L/s	NM	42 (24 - 210)	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kai Tak River in 2016 (Part 1 of 2)

Parameter	Unit	Kai Tak River		
		KN1	KN2	KN3
Dissolved oxygen	mg/L	5.6 (4.3 - 7.2)	6.7 (5.8 - 8.2)	7.0 (6.2 - 8.0)
pH		7.1 (6.9 - 7.6)	7.4 (7.2 - 7.7)	7.4 (7.2 - 7.6)
Suspended solids	mg/L	6 (2 - 13)	10 (3 - 20)	16 (8 - 47)
5-day Biochemical Oxygen Demand	mg/L	5 (1 - 10)	6 (4 - 12)	6 (3 - 18)
Chemical Oxygen Demand	mg/L	25 (13 - 41)	29 (15 - 41)	32 (13 - 41)
Oil & grease	mg/L	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - <0.5)	<0.5 (<0.5 - 0.6)
E. coli	counts/ 100mL	130 000 (38 000 - 430 000)	74 000 (31 000 - 260 000)	73 000 (11 000 - 220 000)
Faecal coliforms	counts/ 100mL	290 000 (83 000 - 790 000)	160 000 (60 000 - 400 000)	180 000 (58 000 - 430 000)
Ammonia-nitrogen	mg/L	1.65 (0.92 - 5.70)	1.40 (0.82 - 5.40)	1.40 (0.72 - 5.00)
Nitrate-nitrogen	mg/L	2.45 (2.10 - 3.50)	3.45 (1.90 - 4.10)	3.25 (2.00 - 4.10)
Total Kjeldahl nitrogen	mg/L	2.60 (1.50 - 6.30)	2.85 (1.60 - 6.20)	2.50 (1.50 - 6.00)
Ortho-phosphate Phosphorus	mg/L	1.15 (0.48 - 1.70)	1.30 (0.36 - 2.00)	1.15 (0.34 - 2.00)
Total phosphorus	mg/L	1.30 (0.55 - 1.90)	1.55 (0.42 - 2.20)	1.55 (0.41 - 2.30)
Total sulphide	mg/L	<0.02 (<0.02 - 0.21)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - 0.02)
Aluminium	µg/L	61 (<50 - 149)	84 (64 - 256)	102 (62 - 301)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 1)	<1 (<1 - 1)	<1 (<1 - 1)
Copper	µg/L	4 (3 - 7)	5 (4 - 22)	7 (4 - 10)
Lead	µg/L	<1 (<1 - 2)	1 (<1 - 4)	1 (<1 - 4)
Zinc	µg/L	21 (15 - 31)	27 (18 - 34)	25 (20 - 41)
Flow	L/s	NM	NM	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Summary of water quality monitoring data for Kai Tak River in 2016 (Part 2 of 2)

Parameter	Unit	Kai Tak River		
		KN4	KN5	KN7
Dissolved oxygen	mg/L	6.8 (5.9 - 8.2)	7.1 (6.4 - 8.3)	7.1 (6.4 - 8.6)
pH		7.5 (7.2 - 7.9)	7.4 (7.1 - 7.6)	7.4 (7.1 - 7.6)
Suspended solids	mg/L	15 (4 - 38)	15 (4 - 46)	11 (3 - 16)
5-day Biochemical Oxygen Demand	mg/L	6 (1 - 18)	7 (2 - 16)	7 (4 - 17)
Chemical Oxygen Demand	mg/L	27 (6 - 42)	30 (12 - 41)	29 (16 - 42)
Oil & grease	mg/L	<0.5 (<0.5 - 0.6)	<0.5 (<0.5 - 0.5)	<0.5 (<0.5 - 0.5)
E. coli	counts/ 100mL	41 000 (6 300 - 220 000)	30 000 (13 000 - 84 000)	6 500 (1 000 - 180 000)
Faecal coliforms	counts/ 100mL	110 000 (23 000 - 370 000)	71 000 (18 000 - 160 000)	15 000 (2 000 - 250 000)
Ammonia-nitrogen	mg/L	1.12 (0.28 - 5.60)	1.02 (0.50 - 5.30)	0.84 (0.36 - 5.30)
Nitrate-nitrogen	mg/L	3.05 (0.71 - 4.10)	3.20 (1.90 - 4.10)	3.40 (3.00 - 4.40)
Total Kjeldahl nitrogen	mg/L	2.25 (0.67 - 6.40)	2.10 (1.10 - 6.80)	1.95 (1.20 - 6.80)
Ortho-phosphate Phosphorus	mg/L	0.85 (0.09 - 2.10)	0.87 (0.20 - 2.10)	0.86 (0.33 - 2.00)
Total phosphorus	mg/L	1.10 (0.15 - 2.60)	1.25 (0.27 - 2.40)	1.15 (0.44 - 2.50)
Total sulphide	mg/L	<0.02 (<0.02 - 0.02)	<0.02 (<0.02 - <0.02)	<0.02 (<0.02 - <0.02)
Aluminium	µg/L	92 (55 - 438)	89 (<50 - 304)	55 (<50 - 141)
Cadmium	µg/L	<0.1 (<0.1 - <0.1)	<0.1 (<0.1 - 0.1)	<0.1 (<0.1 - <0.1)
Chromium	µg/L	<1 (<1 - 2)	<1 (<1 - <1)	<1 (<1 - <1)
Copper	µg/L	6 (3 - 9)	7 (4 - 13)	7 (3 - 9)
Lead	µg/L	<1 (<1 - 6)	<1 (<1 - 5)	<1 (<1 - 1)
Zinc	µg/L	24 (14 - 57)	29 (23 - 54)	25 (18 - 31)
Flow	L/s	NM	NM	NM

Notes:
1. Data presented are in annual medians of monthly samples; except those for faecal coliforms and *E. coli* which are in annual geometric means.
2. Figures in brackets are annual ranges.
3. NM indicates no measurement taken.
4. Values at or below laboratory reporting limits are presented as laboratory reporting limits (see Appendix B).
5. Equal values for annual medians (or geometric means) and ranges indicate that all data are the same as or below laboratory reporting limits.

Compliance with the river Water Quality Objectives in 2016

Watercourse	pH	5-Day Biochemical Oxygen Demand	Chemical Oxygen Demand	Dissolved Oxygen	Suspended Solids	Overall Compliance
Eastern New Territories						
Shing Mun River	76%	93%	98%	100%	100%	**93%**
Lam Tsuen River	100%	89%	98%	100%	100%	**97%**
Tai Po River	100%	100%	100%	100%	100%	**100%**
Tai Po Kau Stream	100%	100%	100%	100%	100%	**100%**
Shan Liu Stream	100%	100%	100%	100%	100%	**100%**
Tung Tze Stream	100%	83%	92%	100%	100%	**95%**
Ho Chung River	100%	96%	100%	100%	100%	**99%**
Sha Kok Mei Stream	100%	100%	100%	100%	100%	**100%**
Tai Chung Hau Stream	96%	92%	100%	100%	100%	**98%**
Tseng Lan Shue Stream	100%	61%	97%	100%	100%	**92%**
Northwestern New Territories						
River Indus	94%	56%	81%	89%	100%	**84%**
River Beas	100%	61%	92%	97%	100%	**90%**
River Ganges	100%	75%	81%	100%	100%	**91%**
Yuen Long Creek	100%	0%	31%	71%	75%	**55%**
Kam Tin River	100%	0%	38%	92%	100%	**66%**
Tin Shui Wai Nullah	100%	67%	100%	96%	100%	**93%**
Fairview Park Nullah	100%	50%	75%	100%	0%	**65%**
Ha Pak Nai Stream	100%	100%	100%	100%	100%	**100%**
Tai Shui Hang Stream	100%	100%	100%	100%	100%	**100%**
Pak Nai Stream	100%	100%	100%	100%	100%	**100%**
Sheung Pak Nai Stream	100%	100%	100%	100%	100%	**100%**
Ngau Hom Sha Stream	100%	100%	100%	100%	100%	**100%**
Tsang Kok Stream	100%	100%	100%	100%	100%	**100%**
Lantau Island						
Mui Wo River	100%	100%	100%	100%	100%	**100%**
Tung Chung River	100%	94%	100%	100%	100%	**99%**
Southwestern New Territories & Kowloon						
Tuen Mun River	99%	71%	97%	97%	100%	**93%**
Pai Min Kok Stream	100%	100%	100%	100%	100%	**100%**
Kau Wa Keng Stream	100%	67%	100%	100%	100%	**93%**
Sam Dip Tam Stream	100%	100%	100%	100%	100%	**100%**
Kai Tak River	100%	35%	60%	100%	100%	**79%**
Average Compliance (All monitoring stations)	**97%**	**76%**	**89%**	**98%**	**98%**	**91%**

Water Quality Index (WQI) for inland waters of Hong Kong

The Water Quality Index (WQI) is a numerical value used to denote, based on monitoring data, the general ecological health of a river. It is relevant to conserving the primary beneficial use of rivers for maintenance of aquatic life. The WQI is calculated based on three key parameters: dissolved oxygen, 5-day biochemical oxygen demand (BOD_5) and ammonia-nitrogen content. The measurements of these parameters are evaluated using the following table:

Calculation of Water Quality Index

No. of points awarded	Dissolved oxygen (% saturation)	BOD_5 (mg/L)	Ammonia-nitrogen (mg/L)
1	91 – 110	< 3	< 0.5
2	71 – 90 111 – 120	3.1 – 6.0	0.5 – 1.0
3	51 – 70 121 – 130	6.1 – 9.0	1.1 – 2.0
4	31 – 50	9.1 – 15.0	2.1 – 5.0
5	< 30 or > 130	> 15.0	> 5.0

All parameters carry equal weighting and the monthly index is the sum of points of the three parameters. The annual WQI of a river station is the average of all monthly WQI in the year. The WQI values range from 3 to 15, corresponding to the following water quality conditions:

Water Quality Index gradings

WQI	Water quality condition
3.0 – 4.5	Excellent
4.6 – 7.5	Good
7.6 – 10.5	Fair
10.6 – 13.5	Bad
13.6 – 15.0	Very Bad

Summary of water quality improvements and pollution loading of major rivers

River	Water Quality Improvement and Pollution Loading
Shing Mun River	As a major river which runs through Sha Tin District, Shing Mun River showed marked improvement over the last decade. The water quality of the Shing Mun River main channel has been graded "Excellent" in 2008 - 2016. The 5-day Biochemical Oxygen Demand (BOD_5) level ranged from 1 - 4 mg/L, i.e. 100% compliance with BOD_5 WQO in 2016. Under the Livestock Waste Control Scheme, all livestock farms in Sha Tin have ceased operation. All factory discharges are controlled under the Water Pollution Control Ordinance. All domestic, commercial and industrial buildings in the urbanised town centre are served by government sewers. Sewage is transferred to the Sha Tin Sewage Treatment Works for secondary treatment and disinfection. Some 30 villages have been connected to public sewerage system. Sewerage construction for another 7 villages is being planned and implemented. A total of 18 Dry Weather Flow Interceptors (DWFIs) have been installed in the catchment. The non-point source loading due to a residential population of 460 000 people and various activities of the non-residential population of 280 000 people in the catchment area was estimated to be 160 kg BOD_5/day. Loading from around 22 000 unsewered population was estimated to be 86 kg BOD_5/day, contributing about 35% of the total loading entering Shing Mun River.
Lam Tsuen River	Lam Tsuen River is a major river which passes through the urban area of Tai Po. The water quality of its downstream station has been graded "Good" in 2000 - 2016. The water quality of Tai Po River, as a major tributary of Lam Tsuen River, has been graded "Excellent" in 2007 – 2016. The BOD_5 level of the downstream stations ranged from 1 – 5 mg/L, i.e. 100% compliance with the BOD_5 WQO in 2016. Under the Livestock Waste Control Scheme, all livestock farms in Tai Po have ceased operation. Factory discharges are controlled under the Water Pollution Control Ordinance. All domestic, commercial and industrial buildings in the urbanised town centres are served by government sewers. Sewage is transferred to the Tai Po Sewage Treatment Works for secondary treatment and disinfection. Sewerage works for villages are being planned and implemented progressively. The non-point source loading due to the residential population 200 000 of people and various activities of non-residential population 70 000 of people in the catchment area was estimated to be 88 kg BOD_5/day. Loading from some 44 000 unsewered population was estimated to be 317 kg BOD_5/day, contributing about 78% of the total loading entering Lam Tsuen River.
Tuen Mun River	Tuen Mun River is a major river which runs across Tuen Mun District. Following the completion of a DWFI at the West Rail Siu Hong Station to collect the upstream polluted water to the Pillar Point Sewage Treatment Works (PPSTW) for treatment and disposal at the sea, and the gradual implementation of village sewerage to connect the village houses in the upper catchment to public sewers, the water quality of Tuen Mun River has shown marked improvement in the last two decades. Its midstream and downstream has been improved and maintained at "Good" to "Excellent" level in 2002 - 2016. The midstream and downstream stations have a BOD_5 level ranged from 1 – 10 mg/L, achieving 92% compliance with BOD_5 WQO. All factory discharges are now controlled under the Water Pollution Control Ordinance. All domestic, commercial and industrial buildings in urban areas

	are served by public sewers. The PPSTW was upgraded to chemically enhanced primary treatment level with UV disinfection in May 2014. A new trunk sewer to the west of Tuen Mun River was completed in 2015 and over 90% of the residential population of 380 000 people in the catchment area are now sewered. Of the some 4 300 unsewered villages houses and 300 squatters, 90% was located in upper reaches of Tuen Mun River. Sewer connection work for the village houses is in progress. In addition to the aforesaid DWFI at upper stream, there is a DWFI installed in the midstream to collect the residual pollutants from the unsewered areas in the eastern catchment of Tuen Mun River. The total loading entering Tuen Mun River arising from the unsewered population of 37 000 people was estimated to be 155 kg BOD_5 per day, contributing about 55% BOD_5 loading of the river.
Yuen Long Creek	As a major river in Yuen Long District, Yuen Long Creek passes through rural areas as well as the densely populated Yuen Long new town and Yuen Long Kau Hui. In 2016, its two upstream stations recorded “Fair” and “Bad” grading, respectively. As a major tributary of Yuen Long Creek, Kam Tin River was graded “Bad” in 2016. The downstream stations of Yuen Long Creek and Kam Tin River had BOD_5 level ranged from 12-170 mg/L and 4-51 mg/L, both exceeding the BOD_5 WQO. Under the Livestock Waste Control Scheme, most of the farms had ceased operation. The remaining 34 livestock farms contributed about 3 400 kg BOD_5/day, 62% BOD_5 loading in the river. Factory discharges are controlled under the Water Pollution Control Ordinance. Of the total residential population of 240 000 people, 55% of them are living in the domestic buildings in the urbanised town centre. Being served by government sewers as well as the San Wai and Yuen Long Sewage Treatment Works. Trunk sewers, pumping stations and village sewerage are being planned and constructed progressively in the catchment. Some 21 000 village houses are still using septic tank system. The loading from the unsewered population of 110 000 people, contributed about 24% BOD_5 loading in the river.
River Indus	River Indus is a major river in the North District. It runs through semi-rural areas like Lung Yeuk Tau, collects runoff from the densely populated Fanling and Sheung Shui urban areas, and meets with River Beas before entering Shenzhen River. The downstream station of River Indus was graded “Fair” to “Good” in 2012 - 2016, with BOD_5 level ranged from 3 – 20 mg/L, i.e. 8% compliance with the BOD_5 WQO in 2016. River Beas, as a major tributary of River Indus, was graded “Good” and had BOD_5 level ranged from 2 - 11 mg/L, i.e. 42% compliance with BOD_5 WQO in 2016 for its downstream station. All livestock farms in the Fanling and Sheung Shui urban areas had ceased operation under the Livestock Waste Control Scheme. There are one pig farm and one poultry farm remaining in the rural areas of River Beas, contributing about 18 kg BOD_5/day. Factory discharges are controlled under the Water Pollution Control Ordinance. Majority of the domestic and commercial buildings in these two urbanised town centres, with a total residential population of 340 000 people, are served by government sewers. Sewage is collected and transported to the Shek Wu Hui Sewage Treatment Works for secondary treatment and disinfection. Sewerage works for some 34 villages are being planned and implemented progressively. Ten DWFI have been installed to collect contaminated runoff from unsewered population. The remaining unsewered population of 88 000 people contributed some 800 kg BOD_5/day, or 52% BOD_5 loading in the river.